

Groups of multiplicative type: homomorphisms into a group scheme

SGA 3, Exposé IX

English draft

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About this draft

This is an unofficial English translation of Exposé IX, “Groups of multiplicative type: homomorphisms into a group scheme” from *Schémas en groupes* (SGA 3). The source is the recomposition edited by P. Gille and P. Polo, PDF `Exp9-8nov09.pdf` (32 pages), from <https://webusers.imj-prg.fr/~patrick.polo/SGA3/>. The seminar was directed by M. Demazure and A. Grothendieck (IHÉS, 1962–1964).

Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a `% typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked `% typo?:`. Scholarly proofreading remains outstanding.

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Exposé IX. Groups of multiplicative type: Homomorphisms into a group scheme

by A. Grothendieck

1. Definitions

Definition 1.1. Let S be a prescheme, G an S -group prescheme. One says that G is a *group of multiplicative type* if G is locally diagonalizable in the sense of the faithfully flat quasi-compact topology (cf. IV 6.3), i.e. if for every $s \in S$ there exists an open neighborhood U of s and a faithfully flat and quasi-compact morphism $S' \rightarrow U$ such that $G' = G \times_S S'$ is a diagonalizable S' -group (I 4.4 and VIII 1.1).

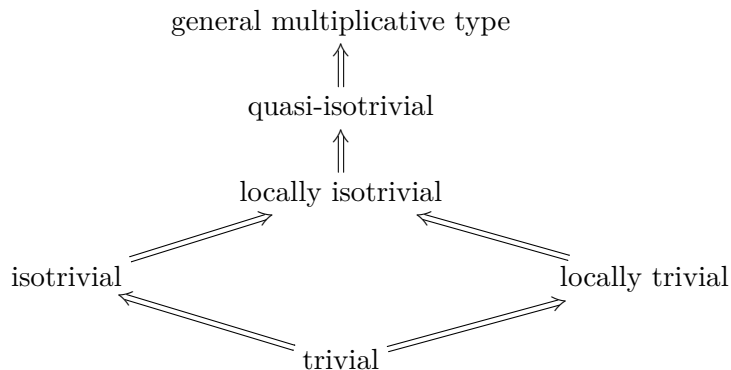
- One says that G is of *quasi-isotrivial* multiplicative type if it is even locally diagonalizable in the sense of the étale topology (IV 6.3), i.e. if, in the preceding definition, one may even take $S' \rightarrow U$ étale surjective, or again (which amounts to the same thing, as one sees by taking the sum scheme of the S' attached to the various $s \in S$) if there exists $S' \rightarrow S$ étale and surjective such that $G' = G \times_S S'$ is a locally diagonalizable S' -group.
- If one may even choose $S' \rightarrow S$ étale surjective and finite, one says that G is of *isotrivial* multiplicative type.
- Finally, one says that G is a group of *locally trivial* (resp. *locally isotrivial*) multiplicative type if every $s \in S$ admits an open neighborhood U such that $G \times_S U$ is a diagonalizable U -group (resp. a group of isotrivial multiplicative type, i.e. there exists $S' \rightarrow U$ a finite surjective étale morphism such that $G \times_S S'$ is a diagonalizable S' -group).

Comments 1.2. One will note that the preceding five notions are all deduced from the notion of a diagonalizable group by the process of localization, in the sense of five different “topologies” associated with (Sch).

One agrees in general that when the word “locally” is not made explicit by specifying the topology considered, it refers to the Zariski topology, in conformity with the received terminology. In the terminology introduced here, “of locally trivial multiplicative type” is equivalent to “locally diagonalizable” of VIII 1.1; similarly, “trivial” translates as “diagonalizable”. Between the five notions introduced, one has the following diagram of implications,

⁽⁰⁾Version 1.0 of November 8, 2009: additions in the proof of 3.6 bis, 4.4–7, 5.0–6, 6.1, 7.1, 8.2. Review 8.1 and 4.5.

which follows from the corresponding relations between the topologies involved:



From the practical point of view, let us point out immediately that all the groups of multiplicative type which we shall encounter will be quasi-isotrivial; thus we shall see in the next exposé (X 4.5) that when G is of finite type over S , G is automatically quasi-isotrivial, but we shall give examples where it is not locally isotrivial. We shall likewise see there that G may be locally trivial without being isotrivial or a fortiori trivial (which easily implies that the inclusions in the preceding diagram are strict).

On the other hand, we shall also see there (X 5.16) that when S is locally noetherian and normal (more generally geometrically unibranch), every group of multiplicative type of finite type over S is necessarily isotrivial, and moreover trivial as soon as it is locally trivial (or merely trivial over a dense open subset). This explains why most of the groups of multiplicative type which one will encounter in practice will doubtless be isotrivial, all the more so since we shall see later that the maximal tori of semisimple group schemes are automatically isotrivial.

Definition 1.3. Let S, G be as in 1.1. One says that G is a *torus* if it is locally isomorphic, in the sense of the faithfully flat quasi-compact topology, to a group of the form $\mathbf{G}_m^r$ (where r is an integer ≥ 0).

With the notation of 1.1, this therefore means that one may choose S' such that G' is isomorphic to a group of the form $(\mathbf{G}_{m,S'})^r$. One will note that the integer r depends on $s \in S$; it is the dimension of the fiber $G_s = G \otimes_S \kappa(s)$. It is a locally constant function of s , as one verifies immediately. There is occasion to generalize this remark:

Definition 1.4. Let G be a diagonalizable group scheme over a field k ; thus G is isomorphic to a group $D_k(M)$, where M is an ordinary commutative group, defined up to isomorphism by this condition, more precisely $M \simeq \text{Hom}_{k\text{-gr.}}(G, \mathbf{G}_{m,k})$ (VIII 1.3). The isomorphism class of M is called the *type* of the diagonalizable group G ; it is evidently invariant under extension of the base field.

If now G is a group scheme of multiplicative type over an arbitrary prescheme S , then for every $s \in S$ there exists an extension k of $\kappa(s)$ such that $G \times_S \text{Spec}(k)$ is a diagonalizable k -group;⁽¹⁾ its type is then independent of the chosen extension by the preceding remark, and will be called the *type of G at s* , or *type of G_s* . In particular, if G itself is diagonalizable, hence isomorphic to a group of the form $D_S(M)$, then for every $s \in S$ the type of G at s equals the class of the group M .

In general, if G is a group of multiplicative type over S , M an ordinary commutative group, one shall say that G is *of type M* if it is of type M at every point $s \in S$, in other words if G is locally isomorphic to $D_S(M)$ in the sense of the faithfully flat and quasi-compact topology.

⁽¹⁾**N.D.E.** Indeed, by hypothesis there exists an open neighborhood U of s and a faithfully flat and quasi-compact morphism $S' \rightarrow U$ such that $G_{S'}$ is diagonalizable; then for every $s' \in S'$ over s , $\kappa(s')$ is an extension of $\kappa(s)$ and $G_{s'} = G \times_S \text{Spec}(\kappa(s'))$ is diagonalizable.

Remark 1.4.1. ⁽²⁾ a) One sees immediately that for a given group of multiplicative type G over S , the function $s \mapsto \text{type of } G_s$ is locally constant on S : indeed, with the notation of 1.1, if G' is of type M , then G is of type M on the neighborhood U of s . It follows that one has a canonical partition of S as a sum of preschemes S_i , such that for every i , G_{S_i} is of type M_i , where the M_i are pairwise nonisomorphic commutative groups.

b) In particular, if S is connected, the type of the fibers of G is constant, i.e. there exists a commutative group M such that G is of type M . Finally, if G is a torus, the type of G at s is characterized by the integer $\dim G_s$ (indeed, G_s is of type $\mathbf{Z}^n$, where $n = \dim G_s$).

Remark 1.5. a) It is trivial from definitions 1.1 and 1.3 that they are “compatible with extension of the base”. Thus, if G is a group prescheme over S and $S' \rightarrow S$ a base change morphism, then if G is of multiplicative type (resp. of isotrivial multiplicative type, etc.), the same is true of G' .

b) When in addition $S' \rightarrow S$ is faithfully flat and quasi-compact, then if G' is of multiplicative type, resp. a torus, the same is true of G . If in addition $S' \rightarrow S$ is étale (resp. finite étale), and G' quasi-isotrivial (resp. isotrivial), the same is true of G .

c) Finally, returning to an arbitrary base change morphism $f : S' \rightarrow S$, if $s' \in S'$ and $s = f(s')$, then the type of G' at s' equals that of G at s , since $G'_{s'} = G_s \otimes_{\kappa(s)} \kappa(s')$.

2. Extension of certain properties of diagonalizable groups to groups of multiplicative type

The extensions in question are essentially trivial consequences of the results of the preceding exposé, in view of definitions 1.1 and the “local” nature (in the sense of the faithfully flat and quasi-compact topology) of the results in question.

Definition 2.0. ⁽³⁾ Denote by $\mathcal{M}$ the set of faithfully flat and quasi-compact morphisms.

Proposition 2.1. *Let G be a group of multiplicative type over a prescheme S . One has the following:*

- a) G is faithfully flat over S , and affine over S (a fortiori quasi-compact over S).
- b) G of finite type over $S \iff G$ of finite presentation over $S \iff$ for every $s \in S$, the type of G at s is given by a commutative group of finite type.
- c) G finite over $S \iff$ for every $s \in S$, the type of G at s is given by a finite commutative group $\iff$ (if S quasi-compact) G is of finite type over S and is annihilated by an integer $n > 0$.
- c') G integral over $S \iff$ for every $s \in S$, the type of G at s is given by a commutative torsion group.
- d) G is the unit group $\iff$ for every $s \in S$, the type of G at s is given by the zero commutative group.
- e) G is smooth over $S \iff$ for every $s \in S$, the type of G at s is given by a commutative group of finite type whose torsion subgroup has order prime to the characteristic of $\kappa(s)$.

These statements follow from VIII 2.1, in view of the fact that the properties in question descend by faithfully flat and quasi-compact morphisms (cf. SGA 1, VIII or EGA IV₂, §2).

Using VIII 3.5, one likewise obtains:

Proposition 2.2. *Let G be a group of multiplicative type and of finite type over S ; then for every integer $n \neq 0$, the kernel ${}_nG$ of $n \cdot \text{id}_G$ is a group of multiplicative type finite over S .*

Proposition 2.3. *Let G be a group of multiplicative type over the prescheme S , acting freely on the right on the S -prescheme X affine over S . Then:*

⁽²⁾**N.D.E.** The following remarks have been given the number 1.4.1, for later references.

⁽³⁾**N.D.E.** This definition has been added; it appears in propositions 2.3 and 2.7.

a) The equivalence relation defined by G in X is $\mathcal{M}$ -effective (IV 3.4); moreover $Y = X/G$ is affine over S .

b) If in addition X is of finite presentation (resp. of finite type) over S , the same is true of Y .

The first assertion follows from VIII 5.1, treating the case where G is diagonalizable, and from IV 3.5.2, which allows one to reduce to it, taking account of the fact that faithfully flat and quasi-compact morphisms $S' \rightarrow S$ are morphisms of effective descent for the fibered category of schemes affine over others, i.e. for every Y' affine over S' , endowed with a descent datum relative to $S' \rightarrow S$, this descent datum is effective, i.e. Y' comes from a Y affine over S , cf. SGA 1, VIII 2.1.

For the second assertion one is also reduced to the diagonalizable case VIII 5.8, since the finiteness conditions considered descend by faithfully flat and quasi-compact morphisms (SGA 1, VIII 3.3 and 3.6⁽⁴⁾). Proceeding as in VIII, corollaries 5.5 to 5.7, one draws from 2.3:

Corollary 2.4. *Under the conditions of 2.3, the graph morphism*

$$X \times_S G \rightarrow X \times_S X$$

is a closed immersion. For every section σ of X over S , the corresponding morphism $G \rightarrow X$, $g \mapsto \sigma \cdot g$, is a closed immersion.

In particular:

Corollary 2.5. *Let $u : G \rightarrow H$ be a monomorphism of S -group preschemes, with G of multiplicative type and H affine over S . Then u is a closed immersion, $H/G = Y$ exists and is affine over S ; finally, H is a principal homogeneous bundle over Y with group G_Y .*

Remark 2.6. Let $u : G \rightarrow H$ be a monomorphism of S -group preschemes, G being of multiplicative type and of finite type over S , H being of finite presentation over S and separated over S . Then one can show that u is a closed immersion, using VIII 7.12 (see remark VIII 7.13 b)).

Proposition 2.7. *Let $u : G \rightarrow H$ be a homomorphism of S -groups of multiplicative type, with H of finite type over S . Then:*

(i) $G' = \text{Ker } u$ is an S -group of multiplicative type; the equivalence relation defined by G' in G is $\mathcal{M}$ -effective, so (IV 3.4) u factors as

$$G \rightarrow G/G' = I \rightarrow H,$$

where $I \rightarrow H$ is a monomorphism⁽⁵⁾ of S -groups; I is of multiplicative type, the equivalence relation in H defined by I is $\mathcal{M}$ -effective, so $H' = H/I$ exists, and moreover H' is of multiplicative type.

(ii) H' and I are of finite type, and the same is true of G' if G is so.

PROOF. Proceeding as for 2.3, one is reduced to the case where G and H are diagonalizable, and then 2.7 reduces to VIII 3.4. $\square$

Let us note the following corollaries:

Corollary 2.8. a) *Let S be a prescheme. Then the category of S -groups of multiplicative type and of finite type is an abelian category.*

b) *Let $u : G \rightarrow H$ be a homomorphism in this category; for it to be a monomorphism (resp. an epimorphism, resp. an isomorphism) in this category, it is necessary and sufficient that u be a monomorphism of preschemes (resp. that u be faithfully flat, resp. that u be an isomorphism of preschemes).*

⁽⁴⁾**N.D.E.** Cf. also V, 9.1 or EGA IV₂, 2.7.1.

⁽⁵⁾**N.D.E.** And even a closed immersion.

It suffices to note that the product of two S -groups of multiplicative type is again of multiplicative type (which is immediate), evidently of finite type over S if both factors are so. The rest of 2.8 follows immediately from 2.7; the details are left to the reader.

Corollary 2.9. *Let $u : G \rightarrow H$ be a homomorphism of S -groups of multiplicative type and of finite type. Let U be the set of points $s \in S$ such that $u_s : G_s \rightarrow H_s$ is a monomorphism (resp. faithfully flat, resp. an isomorphism). Then U is both open and closed, and the induced homomorphism $u_U : G_U \rightarrow H_U$ is a monomorphism (resp. faithfully flat, resp. an isomorphism).*

Let P (resp. Q) be the kernel (resp. cokernel) of u . Thanks to 2.7, Q exists and P and Q are of multiplicative type; moreover formation of P and Q commutes with every base change $S' \rightarrow S$, in particular with formation of the fibers. On the other hand u is a monomorphism (resp. faithfully flat, resp. an isomorphism) if and only if $P = 0$ (resp. $Q = 0$, resp. $P = Q = 0$). Thanks to these remarks one is therefore reduced to checking this: if R is an S -group of multiplicative type, then the set U of $s \in S$ such that $R_s = 0$ is open and closed, and $R_U = 0$. Now this is contained in remark 1.4.1 a).

Corollary 2.10. *Let G be an S -group of multiplicative type and of finite type, H and H' two subgroups of multiplicative type.*

- a) $H'' = H \cap H'$ is a subgroup of multiplicative type of G .
- b) Let U be the set of $s \in S$ such that $H_s \subseteq H'_s$ (resp. $H_s = H'_s$). Then U is open and closed, and $H_U \subseteq H'_U$ (resp. $H_U = H'_U$).

Of course, the intersection sign in $H \cap H'$ denotes intersection in the functorial sense, i.e. $H \times_G H'$, which is evidently a sub- S -group of G ; similarly the inclusion (resp. equality) sign denotes the order relation (resp. equality) between subfunctors of H (and not inclusion (resp. equality) of the underlying sets).

First applying 2.7 to the inclusion morphism $H \rightarrow G$, one finds that H is of finite type; similarly H' is of finite type. It then follows from 2.8 that $H \cap H'$ is of multiplicative type and of finite type (one will note that the canonical functor from the category considered in 2.8 to the category $(\text{Sch})_S$ commutes with finite projective limits).

Formation of $H \cap H'$ commutes with extension of the base, in particular with formation of the fibers. On the other hand $H \subseteq H'$ (resp. $H = H'$) is equivalent to $H'' = H$ (resp. $H'' = H$ and $H'' = H'$). Consider then the canonical homomorphisms $H'' \rightarrow H$ and $H'' \rightarrow H'$; then U is the set of $s \in S$ such that the induced homomorphism $H''_s \rightarrow H_s$ is an isomorphism (resp. such that $H''_s \rightarrow H_s$ and $H''_s \rightarrow H'_s$ are isomorphisms). By 2.9, it follows that U is open and closed, and that $H''_U \rightarrow H_U$ (resp. $H''_U \rightarrow H_U$ and $H''_U \rightarrow H'_U$) are isomorphisms. Whence the desired conclusion.

Proposition 2.11. *Let S be a prescheme, G an S -group of multiplicative type and of finite type over S , H a subgroup of multiplicative type of G , and $K = G/H$ (which is a quotient group of multiplicative type of G).*

(i) *Suppose G trivial, i.e. diagonalizable. Then there exists a partition of S into open and closed subsets S_i such that for every i , H_{S_i} and K_{S_i} are diagonalizable. In particular, if S is connected, H and K are diagonalizable.*

(ii) *Same statement as in (i), replacing “diagonalizable” by “isotrivial”, provided S is connected, or locally connected, or quasi-compact.*

(iii) *Suppose G locally trivial (resp. locally isotrivial, resp. quasi-isotrivial); then the same is true of K and H .*

PROOF. (i) By hypothesis one has $G = D_S(M)$, where M is a commutative group of finite type. For every quotient group M_i of M , let $H_i = D_S(M_i)$ be the corresponding diagonalizable subgroup of G (VIII 3.1). Let S_i be the set of $s \in S$ such that $H_s = (H_i)_s$; by 2.10, S_i is open and closed, and one has $H_{S_i} = (H_i)_{S_i}$, so H_{S_i} is diagonalizable, and hence so

is K_{S_i} (VIII 3.1). Evidently the S_i are pairwise disjoint; I say that they cover S . This follows from the fact that for every s , H_s is diagonalizable, as a subgroup of the diagonalizable group G_s (cf. 8.1 below⁽⁶⁾), and hence H_s is of the form $D_{\kappa(s)}(M_i)$ by VIII, 1.5 and 3.2 b). Restricting to the family of nonempty S_i , conclusion (i) appears.

(ii) By hypothesis, there exists $S' \rightarrow S$ étale finite surjective such that $G_{S'}$ is diagonalizable. Thus every point of S' has a neighborhood U' , both open and closed, such that $H_{U'}$ and $K_{U'}$ are diagonalizable. Then the image U of U' in S is both open and closed, and $S' \rightarrow S$ still induces a finite surjective étale morphism $U' \rightarrow U$; one therefore sees that every point s of S has an open and closed neighborhood U such that H_U and K_U are isotrivial. Conclusion (ii) follows immediately.

(iii) Follows immediately from (i) and (ii) and the definitions. Note also that the “quasi-isotrivial” case will also follow from the more general fact “finite type $\Rightarrow$ quasi-isotrivial”, announced in 1.2 (cf. X 4.5). $\square$

3. Infinitesimal properties: Lifting and conjugacy theorems

The fundamental infinitesimal properties of groups of multiplicative type follow from the following theorem:

Theorem 3.1. *Let S be an affine scheme, H an S -group of multiplicative type, $\mathcal{F}$ a quasi-coherent sheaf over S on which H acts (I 4.7). Then one has*

$$H^i(H, \mathcal{F}) = 0 \quad \text{for } i > 0,$$

where H^i is the Hochschild cohomology studied in Exp. I, §5.

Indeed, by loc. cit., 5.3, Hochschild cohomology is made explicit, in terms of the affine rings A of S and B of G , and of the module M over A defining $\mathcal{F}$, as the cohomology of a complex of A -modules $C^\bullet(H, M)$, whose formation commutes with every base change $A \rightarrow A'$. Consequently, for a base change $A \rightarrow A'$ with A' flat over A , one shall have

$$H^i(H', \mathcal{F}') = H^i(H, \mathcal{F}) \otimes_A A',$$

and therefore, if one even supposes A' faithfully flat over A , to prove that $H^i(H, \mathcal{F})$ is zero it suffices to prove that $H^i(H', \mathcal{F}')$ is so. This remark reduces us to checking 3.1 in the case where G is diagonalizable; in this case it was proved in I 5.3.3. Using the results of Exposé III, we shall draw from 3.1 various consequences of a geometric nature:

Theorem 3.2. *Let S be an affine scheme, S_0 an affine subscheme defined by an ideal $\mathcal{J}$, H a group of multiplicative type over S , G any group prescheme over S , $u, v : H \rightarrow G$ two group homomorphisms, $u_0, v_0 : H_0 \rightarrow G_0$ the homomorphisms deduced from u, v by the base change $S_0 \rightarrow S$. If $\mathcal{J}^2 = 0$ and $u_0 = v_0$, then there exists a $g \in G(S)$ such that $v = \text{int}(g) \circ u$ and $g_0 = e$.*

This follows from III 2.1 (ii) and from 3.1 (vanishing of H^1).

Corollary 3.3. *Under the preliminary conditions of 3.2, suppose in addition G smooth over S , but, on the other hand, suppose only $\mathcal{J}$ nilpotent (not necessarily of square zero). If there exists $g_0 \in G_0(S_0)$ such that $v_0 = \text{int}(g_0) \circ u_0$, then g_0 lifts to a $g \in G(S)$ such that $v = \text{int}(g) \circ u$.*

By induction on the integer n such that $\mathcal{J}^n = 0$, one is reduced to the case where $\mathcal{J}$ has square zero. Moreover, G being smooth over S , one can lift g_0 to an element g' of $G(S)$. Replacing v by $v' = \text{int}(g') \circ u$ if necessary, one is then reduced to situation 3.2.

⁽⁶⁾**N.D.E.** The erroneous reference to IV 4.7.5 has been corrected. Note that no. 8 of the present exposé is independent of nos. 3 to 7.

Corollary 3.4. *Under the preliminary conditions of 3.2, with $\mathcal{J}$ nilpotent, suppose in addition u central (for example G commutative). Then $u_0 = v_0$ implies $u = v$.*

Indeed, reduction to the case of $\mathcal{J}$ of square zero is again immediate, and then it suffices to apply 3.2. In particular:

Corollary 3.5. *Let S, S_0, H, G be as in 3.2, with $\mathcal{J}$ nilpotent. Let $u : H \rightarrow G$ be a homomorphism of S -groups such that $u_0 : H_0 \rightarrow G_0$ is the unit homomorphism; then u is the unit homomorphism.*

Theorem 3.6. *Let S be an affine scheme, S_0 a subscheme defined by a nilpotent ideal $\mathcal{J}$, H a group of multiplicative type over S , G a group prescheme smooth over S , $u_0 : H_0 \rightarrow G_0$ a homomorphism of the S_0 -groups deduced by the base change $S_0 \rightarrow S$.*

Then there exists a homomorphism $u : H \rightarrow G$ of S -groups lifting u_0 (and by 3.3 two such liftings u, u' are conjugate by an element of $G(S)$ reducing to the unit element of $G(S_0)$).

This follows from III 2.1 and 2.3, and from 3.1 (vanishing of H^2 for the existence of the lifting of u_0 , vanishing of H^1 for uniqueness up to conjugacy).

One can similarly prove the following variants of 3.2 to 3.6 (which in fact imply the preceding statements, by applying them to the graph subgroups of the homomorphisms considered in 3.2 to 3.6):

Theorem 3.2 bis. *Let S be an affine scheme, S_0 a subscheme defined by an ideal $\mathcal{J}$, G an S -group prescheme, H, H' subgroup preschemes of multiplicative type, G_0, H_0, H'_0 the groups over S_0 deduced by the base change $S_0 \rightarrow S$. If $\mathcal{J}^2 = 0$ and $H_0 = H'_0$, then there exists $g \in G(S)$ such that $H' = \text{int}(g)(H)$ and $g_0 = e$.*

Corollary 3.3 bis. *Under the preliminary conditions of 3.2 bis, suppose in addition G smooth over S , and $\mathcal{J}$ nilpotent (not necessarily of square zero). Let $g_0 \in G_0(S_0)$ be such that $H'_0 = \text{int}(g_0)(H_0)$; then g_0 lifts to $g \in G(S)$ such that $H' = \text{int}(g)(H)$.*

Corollary 3.4 bis. *Under the preliminary conditions of 3.2 bis, suppose $\mathcal{J}$ nilpotent, H central (for example G commutative). Then $H_0 = H'_0$ implies $H = H'$.*

Theorem 3.6 bis. *Let S be an affine scheme, S_0 a subscheme defined by a nilpotent ideal $\mathcal{J}$, G a smooth S -group prescheme, H_0 a subgroup prescheme of $G_0 = G \times_S S_0$, of multiplicative type. Then:*

- (a) *There exists a subgroup prescheme H of G , flat over S , such that $H \times_S S_0 = H_0$.*
- (b) *Such an H is necessarily of multiplicative type.*
- (c) *Finally, by 3.2 bis, two such liftings H, H' of H_0 are conjugate by an element $g \in G(S)$ reducing to the unit element of $G(S_0)$.*

Let us show how one can prove 3.2 bis (of which 3.3 bis and 3.4 bis are an immediate consequence) and assertion (a) of 3.6 bis. The latter follows from 3.1 (vanishing of H^2) and III 4.37 (ii). Similarly, 3.2 bis follows from 3.1 (vanishing of H^1) and III 4.37 (i), at least when G and H are smooth over S .

But using the results of the next exposé, one can very simply deduce 3.2 bis and 3.6 bis, in the general form stated here, from 3.2 resp. 3.6. For 3.2 bis, it indeed suffices to note that by X 2.1, H and H' are isomorphic, which reduces us to 3.2.

For 3.6 bis, one notes that H_0 is necessarily of finite type over S_0 :⁽⁷⁾ indeed, as a subprescheme of G_0 , which is smooth over S_0 , H_0 is locally of finite type over S_0 ; now by 2.1 a), H_0 is affine over S_0 , and hence of finite type over S_0 . Then, by X 4.5, H_0 is quasi-isotrivial and therefore comes, by X 2.1, from a group of multiplicative type H over S .⁽⁸⁾ Then, by 3.6,

⁽⁷⁾**N.D.E.** The original indicated: “by 2.1 b), since its fibers are so”. The editors did not understand this argument and have substituted the following one for it.

⁽⁸⁾**N.D.E.** The following sentence has been expanded.

there exists a homomorphism of S -groups $u : H \rightarrow G$ lifting the immersion $H_0 \hookrightarrow G_0$; since H and H_0 (resp. G and G_0) have the same underlying topological space, and since for every $h \in H$ the morphism $\mathcal{O}_{G,u(h)} \rightarrow \mathcal{O}_{H,h}$ is surjective (for it is so after reduction modulo $\mathcal{J}$, which is nilpotent), u is also an immersion (cf. EGA I, 4.2.2).

Finally, for every lifting H of H_0 to a flat subgroup of G , H is necessarily of multiplicative type by X 2.3,⁽⁹⁾ which also proves assertion (b) of 3.6 bis. (The reader will check that results 3.2 bis to 3.6 bis are not used in Exposé X, so there is no vicious circle!)

Proposition 3.8. ⁽¹⁰⁾ *Let S be a prescheme, S_0 a closed subprescheme defined by a locally nilpotent ideal $\mathcal{J}$, G an S -group of multiplicative type, X an S -prescheme locally of finite presentation. Suppose that G acts on X in such a way that $G_0 = G \times_S S_0$ acts trivially on $X_0 = X \times_S S_0$. Under these conditions, G acts trivially on X .*

The proof is that of III 2.4 b); one can also reduce to loc. cit. by noting that 3.8 reduces immediately to the case where G is diagonalizable, which is the case considered in loc. cit.

Corollary 3.9. *Let S, S_0 be as above, and $u : G \rightarrow H$ a homomorphism of S -groups, with G of multiplicative type and H locally of finite presentation over S . Suppose that the homomorphism $u_0 : G_0 \rightarrow H_0$ deduced by base change $S_0 \rightarrow S$ is central. Then u is central.*

Indeed, it suffices to apply 3.8 by putting $X = H$ and making G act on H by $(g, h) \mapsto \text{int}(u(g)) \cdot h = u(g)hu(g)^{-1}$.

4. The density theorem

This theorem (cf. 4.7 below),⁽¹¹⁾ together with the theorem⁽¹²⁾ of no. 7, will be the essential tool in the present exposé and the following two for passing from the infinitesimal properties of groups of multiplicative type which we have just developed to the “finite” properties.

Definition 4.1. Let X be a prescheme. One says that a family $(Z^i)_{i \in I}$ of subpreschemes of X is *schematically dense* if for every open subset U of X and every closed subprescheme U' of U containing the $Z^i \cap U$, one has $U' = U$.

One says that a subprescheme Z of X is schematically dense in X if the family consisting of Z is so.

One sees immediately (cf. EGA IV₃, 11.10.1) that the definition is equivalent to saying that for every open subset U of X , every section f of $\mathcal{O}_U$ which is zero on the $Z^i \cap U$ is zero, which also means that the intersection of the kernels of the canonical homomorphisms

$$u_i : \mathcal{O}_X \rightarrow (v_i)_*(\mathcal{O}_{Z^i})$$

is zero, where $v_i : Z^i \rightarrow X$ is the canonical immersion.⁽¹³⁾ When X is over a prescheme S , this is also equivalent to saying that for every open subset U of X and every pair (u, v) of morphisms from U to an S -prescheme Y separated over S , coinciding on the $Z^i \cap U$, one has $u = v$. (Indeed, the relation $u = v$ is equivalent to the relation $U' = U$, where U' is the inverse image of the diagonal of $Y \times_S Y$ by the S -morphism $X \rightarrow Y \times_S Y$ defined by (u, v) ; this diagonal is a closed subprescheme of $Y \times_S Y$, so U' is a closed subprescheme of U containing the $Z^i \cap U$ by the hypothesis on (u, v) , so if the family (Z^i) is schematically dense,

⁽⁹⁾**N.D.E.** One will note that X 2.3 depends essentially on theorem 3.6 of the present exposé.

⁽¹⁰⁾**N.D.E.** The original numbering has been preserved: there is no no. 3.7.

⁽¹¹⁾All results from 4.1 to 4.6 are contained in EGA IV₃, 11.10, to which we refer the reader for a formal exposition of the notion of schematic density.

⁽¹²⁾**N.D.E.** The “algebraicity” theorem 7.1.

⁽¹³⁾**N.D.E.** All the schematic density results stated in EGA IV₃, §11.10 follow from the results on “separating families of homomorphisms” proved in loc. cit., §11.9, to which we shall refer in some of the following N.D.E.

one shall have $U' = U$, whence $u = v$; one sees the converse implication by simply taking $Y = \operatorname{Spec} \mathcal{O}_S[T]$.)

With the terminology introduced in EGA I, end of no. 9.5, saying that the subscheme Z of X is schematically dense also means that X is identical to the closure subscheme of Z in X .

Lemma 4.2. *Let X be a flat S -prescheme, $(Z^i)_{i \in I}$ a family of subschemes of X , flat over S . Let S_0 be a subscheme of S , defined by a nilpotent ideal $\mathcal{J}$; suppose the modules $\mathcal{J}^n/\mathcal{J}^{n+1}$ locally free over S_0 .*

Let X_0, Z_0^i be deduced from X, Z^i by the base change $S_0 \rightarrow S$. Then if the family $(Z_0^i)_{i \in I}$ is schematically dense in X_0 , the family $(Z^i)_{i \in I}$ is so in X .

Suppose $\mathcal{J}^{n+1} = 0$ (where $n \geq 0$), and argue by induction on n , the assertion being trivial for $n = 0$. Defining S_m, X_m, Z_m^i by reduction modulo $\mathcal{J}^{m+1}$ as usual, the induction hypothesis already implies that $(Z_{n-1}^i)_{i \in I}$ is schematically dense in X_{n-1} . Replacing X by an induced open subset if necessary, we are reduced to proving that every section f of $\mathcal{O}_X$ vanishing on the Z^i is zero.

Now the section f_{n-1} of $\mathcal{O}_{X_{n-1}} = \mathcal{O}_X/\mathcal{J}^n \mathcal{O}_X$ defined by f vanishes on the Z_{n-1}^i , and is therefore zero, so f is a section of $\mathcal{J}^n \mathcal{O}_X$. Since X is flat over S , one has

$$\mathcal{J}^n \mathcal{O}_X \xleftarrow{\sim} \mathcal{E} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{X_0}, \quad \text{where } \mathcal{E} = \mathcal{J}^n = \mathcal{J}^n/\mathcal{J}^{n+1}.$$

Similarly, since each Z^i is flat over S , the restriction f_i of f to Z^i may be regarded as a section of:

$$\mathcal{J}^n \mathcal{O}_{Z^i} \xleftarrow{\sim} \mathcal{E} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Z_0^i}.$$

By hypothesis the f_i are zero. Now $\mathcal{E}$ is locally free by hypothesis, and hence so is $\mathcal{F} = \mathcal{E} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{X_0}$. Thus f is a section of the locally free module $\mathcal{F}$ over X_0 , such that for every i its restriction to Z_0^i is zero. Since $(Z_0^i)_{i \in I}$ is schematically dense in X_0 , it follows immediately that f is zero. Q.E.D.

Corollary 4.3. *Let X be a locally noetherian S -prescheme flat over S , $(Z^i)_{i \in I}$ a family of subschemes of X flat over S . Suppose that for every $s \in S$, the family $(Z_s^i)_{i \in I}$ of fibers at s is schematically dense in X_s . Then the family $(Z^i)_{i \in I}$ is schematically dense in X .*

Replacing X by an induced open subset if necessary, one is reduced to proving that every section f of $\mathcal{O}_X$ whose restriction to the Z^i is zero is zero.⁽¹⁴⁾ For this it suffices to prove that for every $x \in X$ the image of f in $\mathcal{O}_{X,x}$ is zero. Denote by $\mathfrak{m}_x$ the maximal ideal of $\mathcal{O}_{X,x}$, and by $\mathfrak{m}_s$ that of $\mathcal{O}_{S,s}$, where s is the image of x in S . As $\mathcal{O}_{X,x}$ is noetherian, one has $\bigcap_{n \in \mathbf{N}} \mathfrak{m}_x^n = 0$, whence, a fortiori, $\bigcap_{n \in \mathbf{N}} \mathcal{O}_{X,x} \mathfrak{m}_s^n = 0$.

Thus it suffices to show that for every integer n the section induced by f on $X \times_S \operatorname{Spec}(\mathcal{O}_{S,s}/\mathfrak{m}_s^{n+1})$ is zero. By hypothesis the family of fibers $Z_s^i = Z^i \otimes_{\mathcal{O}_{S,s}} \kappa(s)$ is schematically dense in the fiber $X_s = X \otimes_{\mathcal{O}_{S,s}} \kappa(s)$. This therefore reduces us to the case where S is the spectrum of a local ring whose maximal ideal J is nilpotent, $S_0 = \operatorname{Spec} \kappa(s)$, and the hypotheses of 4.2 are satisfied, whence the conclusion.

Lemma 4.4. *Let S be a locally noetherian prescheme, X a locally noetherian S -prescheme flat over S , $(Z^i)_{i \in I}$ a family of subschemes of X flat over S . Suppose that for every $s \in S$, the family $(Z_s^i)_{i \in I}$ of fibers at s is schematically dense in X_s .*

Then for every base change $S' \rightarrow S$ (S' not necessarily locally noetherian), the family $(Z_n^i)_{i \in I}$ is schematically dense in X' (i.e. the family $(Z^i)_{i \in I}$ is universally schematically dense in X relative to S , cf. EGA IV₃, Def. 11.10.8).

⁽¹⁴⁾**N.D.E.** The original has been expanded in what follows.

PROOF. ⁽¹⁵⁾ Let f' be a section of $\mathcal{O}_{X'}$ over an open subset U' of X' which vanishes on the $Z_n^{i'}$; one must show that f' is zero. Let $s' \in S'$; it suffices to prove that f' is zero at all points of U' over s' . Let s be the image of s' in S ; replacing S, S' by the spectra of the local rings at s, s' if necessary, one may suppose S, S' local and s, s' their closed points. One may further suppose X affine, hence

$$S = \operatorname{Spec}(A), \quad S' = \operatorname{Spec}(A'), \quad X = \operatorname{Spec}(B), \quad X' = \operatorname{Spec}(B'), \quad B' = B \otimes_A A'$$

where A, A' are local, $A \rightarrow A'$ is a local homomorphism, and A, B are noetherian. One may also suppose U' affine, of the form $X'_{g'}$, with $g' \in B'$.

For every A -subalgebra A'' of A' , one considers $S'' = \operatorname{Spec}(A'')$ and $X'', Z_n^{i''}$ deduced from X, Z^i by the base change $S'' \rightarrow S$, whence the diagram:

$$\begin{array}{ccccc} Z^i & \longleftarrow & Z_n^{i''} & \longleftarrow & Z_n^{i'} \\ \downarrow & & \downarrow & & \downarrow \\ X & \longleftarrow & X'' & \longleftarrow & X' \\ \downarrow & & \downarrow & & \downarrow \\ S & \longleftarrow & S'' & \longleftarrow & S' \end{array}$$

We shall henceforth restrict to those A'' which are localizations of A -subalgebras of finite type A'_1 of A' at $\mathfrak{m}' \cap A''_1$ (where $\mathfrak{m}'$ is the maximal ideal of A'), so that the homomorphisms $A \rightarrow A'' \rightarrow A'$ are local. Note that these A'' form an increasing filtering family of subalgebras whose union is A' , so their inductive limit is also A' . One therefore has similarly $B' = \varinjlim_{A''} B \otimes_A A''$. Thus there exists an A'' and an element $g'' \in B'' = B \otimes_A A''$ whose image in B' is g' .

⁽¹⁶⁾ By lemma 4.5 below, the property that the family of fibers $(Z_s^i)_{i \in I}$ is schematically dense in X_s is preserved by every base change $S'' \rightarrow S$; consequently, since each S'' is locally noetherian, if one replaces S by a suitable S'' and X, Z^i by $X'', Z^{i''}$, the hypotheses of 4.4 will be preserved.

Thus, replacing S by S'' (and X, Z^i by $X'', Z^{i''}$) if necessary, one may suppose that g' comes from $g \in B$. Replacing X by the open subset $X_g = U$ if necessary, one may therefore suppose that $U' = X'$. One sees similarly that one may suppose that f' comes from a section f of $\mathcal{O}_X$ over X .

Let $Y = \mathcal{V}(f)$ be the subscheme of X defined by f , and $Y^i = Z^i \times_X Y$ its intersection with Z^i , which is a closed subscheme of Z^i , equal to $\mathcal{V}(f^i)$, where f^i is the section of $\mathcal{O}_{Z^i}$ induced by f . Denoting by $Y', Y^{i'}$ the S' -schemes deduced from the preceding ones by base change, one shall still have

$$Y' = \mathcal{V}(f'), \quad Y^{i'} = Z^{i'} \times_{X'} Y' = \mathcal{V}(f^{i'}),$$

and one has the analogous relations for $Y'', Y^{i''}$. The hypothesis that f' vanishes on the $Z^{i'}$ is expressed by the relations $Y^{i'} = Z^{i'}$ for every i .⁽¹⁷⁾

Let $\mathfrak{m}$ be the maximal ideal of A . For every integer $n \geq 0$, introduce the subscheme $S_n = \operatorname{Spec}(A/\mathfrak{m}^{n+1})$ of S and the schemes X_n, Y_n, Z_n^i, Y_n^i deduced from X, Y, Z^i, Y^i by the base change $S_n \rightarrow S$. In general, for every S -prescheme P , one shall put $P_n = P \times_S S_n$.

⁽¹⁵⁾**N.D.E.** For another proof, using a reduction to the case where S' is locally noetherian (and 4.3 and 4.5 as here), see EGA IV₃, 11.9.16 and 11.9.12 (N.B. in the last line of the proof of 11.9.16, replace 11.9.5 by 11.9.12).

⁽¹⁶⁾**N.D.E.** The following sentence has been added, and the original expanded in what follows.

⁽¹⁷⁾**N.D.E.** The original added: “at least at every point x' of X' over s' ”, but this restriction seems unnecessary.

For every n and $i \in I$, consider the functor

$$F_n^i = \prod_{Z_n^i/S_n} Y_n^i/Z_n^i : (\text{Sch})_{/S_n}^\circ \longrightarrow (\text{Ens})$$

defined by (cf. VIII, 6.4): for every P over S_n ,

$$F_n^i(P) = \Gamma((Y_n^i)_P/(Z_n^i)_P) = \begin{cases} \emptyset & \text{if } (Y_n^i)_P \neq (Z_n^i)_P; \\ \{\text{id}\} & \text{if } (Y_n^i)_P = (Z_n^i)_P. \end{cases}$$

Since S_n is local artinian and Z_n^i flat, hence essentially free over S_n , by VIII, 6.4, each F_n^i is representable by a closed subscheme T_n^i of S_n .

(18) The completion $\hat{A}$ of the local ring A is noetherian since A is so, and it is the projective limit of the A_n . Denote by K_n^i the ideal of $A_n = A/\mathfrak{m}^{n+1}$ defining T_n^i , and by K^i the projective limit of the K_n^i ; it is an ideal of $\hat{A}$.

For fixed i , $m \geq n$, and every S_n -prescheme P , one has

$$(Y_n^i)_P = Y^i \times_S S_n \times_{S_n} P = Y^i \times_S P = (Y_m^i)_P,$$

and similarly $(Z_n^i)_P = (Z_m^i)_P$; it follows that F_n^i is the restriction $F_m^i \times_{S_m} S_n$ of F_m^i to $(\text{Sch})_{/S_n}$, whence $T_n^i = T_m^i \times_{S_m} S_n$. One therefore has a commutative diagram with exact rows:

$$\begin{array}{ccccccc} 0 & \longrightarrow & K_m^i & \longrightarrow & A_m & \longrightarrow & \mathcal{O}(T_m^i) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & K_n^i & \longrightarrow & A_n & \longrightarrow & \mathcal{O}(T_n^i) \longrightarrow 0 \end{array}$$

where all the vertical arrows are surjective. This has the following consequences: on the one hand, the projective system $(K_n^i)_{n \in \mathbf{N}}$ satisfies the Mittag-Leffler condition, and hence the projective limit of the $\mathcal{O}(T_n^i)$ is identified with the topological quotient ring $\hat{A}/K^i$ (cf. EGA 0III, §13.2). On the other hand, the map $K^i \rightarrow K_n^i$ is surjective (cf. [BEns], III, §7.4, Prop. 5), whence it follows that $K_n^i \simeq (K^i + \mathfrak{m}^{n+1}\hat{A})/\mathfrak{m}^{n+1}\hat{A}$ and $\mathcal{O}(T_n^i) \simeq (\hat{A}/K) \otimes_A (A/\mathfrak{m}^{n+1})$. In other words, $(T_n^i)_n$ is an inductive system of artinian affine schemes, and the inductive limit $T^i = \varinjlim_n T_n^i$ is a closed formal subscheme of the formal scheme $\hat{S} = \text{Spf}(\hat{A})$ (cf. EGA I, §10), whose reduction modulo $\mathfrak{m}^{n+1}$ is T_n^i .

Let T be the closed formal subscheme of $\hat{S}$ which is the intersection of the T^i , i.e. defined by $K = \sum_i K^i$. Since $\hat{A}$ is noetherian, there exists a finite subset J of I such that one has $K = \sum_{i \in J} K^i$ (i.e. $T = \bigcap_{i \in J} T^i$). Let us then note that for every n one has $K_n = \sum_{i \in J} K_n^i$, where K_n denotes the image of K in A_n .

Recall that f^i denotes the image of $f \in B = \mathcal{O}(X)$ in $\mathcal{O}(Z^i)$, and $f^{i'}$ its image in $\mathcal{O}(Z_n^{i'}) = \mathcal{O}(Z^i) \otimes_A A'$; the hypothesis $Y^{i'} = Z^{i'}$ is equivalent to the vanishing of $f^{i'}$. As $\mathcal{O}(Z^i) \otimes_A A'$ is the inductive limit of the subalgebras $\mathcal{O}(Z^i) \otimes_A A''$ (where A'' satisfies the conditions made explicit above), there therefore exists an A'' such that $Y^{i''} = Z^{i''}$. A priori, A'' depends on i , but one can find an A'' which works for all $i \in J$, since J is finite. Put $S'' = \text{Spec}(A'')$ and $S''_{(n)} = S'' \times_S S_n$.⁽¹⁹⁾

Since $Y^{i''} \times_{S''} S''_{(n)} = (Y_n^i)_{S''_{(n)}}$ equals $Z^{i''} \times_{S''} S''_{(n)} = (Z_n^i)_{S''_{(n)}}$ for every $i \in J$, it follows from the definition of the T_n^i that $S''_{(n)} \rightarrow S_n$ factors through T_n^i for every $i \in J$, and hence also through $T_n = \bigcap_{i \in J} T_n^i$, and hence also through T_n^i for every $i \in I$. Denoting by f'' the image of f in $B'' = B \otimes_A A''$, and by $f''_{(n)}$ its image in $B''_{(n)} = B''/\mathfrak{m}^{n+1}B''$, this means that for every n , $f''_{(n)}$ vanishes on the $Z^i \times_S S''_{(n)}$.

(18) **N.D.E.** The original has been expanded and corrected to show that the projective limit of the rings $\mathcal{O}(T_n^i)$ defines a closed formal subscheme of the formal scheme $\text{Spf}(\hat{A})$.

(19) **N.D.E.** The original has been expanded and corrected, distinguishing $S''_{(n)}$ from the subscheme $S''_n = S''(s'')$ introduced below.

Since the morphisms $A \rightarrow A'' \rightarrow A'$ are local, the image s'' of s' in S'' is the closed point of S'' , corresponding to the maximal ideal $\mathfrak{m}''$ of B'' , and the image of s'' in S is s . Fix n and now denote by S''_n the n -th infinitesimal neighborhood of s'' in S'' , i.e. $\text{Spec}(A''/\mathfrak{m}''^{n+1})$. Put $B''_n = B''/\mathfrak{m}''^{n+1}B''$ and $Z''_n = Z^i \times_S S''_n = \text{Spec}(B''_n)$. Then the image f''_n of $f''_{(n)}$ in B''_n vanishes on Z''_n for every i . Now, by lemma 4.5 below, the family of fibers

$$Z''_0 = Z''_n \times_{S''_n} \kappa(s'') = Z^i_s \times_{\kappa(s)} \kappa(s'')$$

is schematically dense in the fiber $X''_0 = X''_n \times_{S''_n} \kappa(s'') = X_s \times_{\kappa(s)} \kappa(s'')$. Thus $S''_n, X''_n, (Z''_n)_{i \in I}$, and $S''_0 = \text{Spec } \kappa(s'')$ satisfy the hypotheses of lemma 4.2; it therefore follows that $f''_n = 0$, i.e. $f'' \in \mathfrak{m}''^{n+1}B''$, for every n . Since B is noetherian and $B'' = B \otimes_A A''$ is a localization of a B -algebra of finite type, B'' is noetherian, and hence its local rings are separated for their usual topology. It follows that f'' is zero at the points x'' of X'' such that $\mathfrak{m}''B''$ is contained in the maximal ideal of $\mathcal{O}_{X'', x''}$, i.e. at the points of X'' over s'' . A fortiori, f' is zero at the points x' of X' over s' . $\square$

Lemma 4.5.0. ⁽²⁰⁾ *Let X be an S -prescheme, $(Z^i)_{i \in I}$ a family of subpreschemes of X , and $S' \rightarrow S$ a faithfully flat morphism. If the family $(Z'^i)_{i \in I}$ is schematically dense in X' , then $(Z^i)_{i \in I}$ is schematically dense in X .*

This follows from EGA IV₃, 11.10.5 (i) and 11.9.10 (i). It remains to set out the proof of the

Lemma 4.5. *Let X be a locally noetherian⁽²¹⁾ prescheme over a field k , $(Z^i)_{i \in I}$ a family of subpreschemes of X , k' an extension of k , X' and Z'^i the preschemes deduced from X and Z^i by the base change k'/k . For $(Z^i)_{i \in I}$ to be schematically dense in X , it is necessary and sufficient that $(Z'^i)_{i \in I}$ be so in X' .*

The “sufficient” part follows from 4.5.0; let us prove the “necessary” part.⁽²²⁾ First one may suppose $X = \text{Spec}(B)$ affine. For every i , let $(Z^{ij})_{j \in J_i}$ be a covering of Z^i by affine open subsets; replacing $(Z^i)_{i \in I}$ by the family $(Z^{ij})_{(i,j) \in J}$, where $J = \coprod_{i \in I} J_i$, one may also suppose the Z^i affine.

Let $g \in B' = B \otimes_k k'$ and $t' \in B'_g$ be a section of $\mathcal{O}_{X'}$ on the affine open subset $U' = X'_g$, zero on the $Z'^i \cap U'$, i.e. whose image in each $(\mathcal{O}(Z^i) \otimes_k k')_g$ is zero. There exists a k -subalgebra of finite type A of k' such that $g \in B_A = B \otimes_k A$ and $t' \in (B_A)_g$. Since the map $\mathcal{O}(Z^i) \otimes_k A \rightarrow \mathcal{O}(Z^i) \otimes_k k'$ is injective, the same is true of the map

$$(\mathcal{O}(Z^i) \otimes_k A)_g \rightarrow (\mathcal{O}(Z^i) \otimes_k k')_g;$$

in view of the hypothesis, this implies that the image of t' in each $(\mathcal{O}(Z^i) \otimes_k A)_g$ is zero. This reduces us to showing that the family $(Z^i_A)_{i \in I}$ is schematically dense in X_A ,⁽²³⁾ which follows from EGA IV₃, 11.9.10 b).

Let us note in passing the following result, which will serve in a later exposé:⁽²⁴⁾

Corollary 4.6. *Let X be a flat S -prescheme, U an open subset of X . Suppose that for every $s \in S$, U_s is schematically dense in X_s , and that X is locally noetherian, or locally of finite presentation over S . Then U is schematically dense in X .*

⁽²⁰⁾**N.D.E.** This lemma has been inserted here; it is used in the proof of 4.5 and 4.7.

⁽²¹⁾**N.D.E.** This hypothesis is in fact superfluous, cf. EGA IV₃, 11.10.6 and 11.9.13.

⁽²²⁾**N.D.E.** The original has been expanded in what follows.

⁽²³⁾**N.D.E.** The original continued: “Using 4.3 for the $X_{A'}$, where A' is a quotient of A by a power of a maximal ideal, and Hilbert’s Nullstellensatz, one is reduced to the case where k' is a finite extension of k ”. One could expand this reduction and proceed along these lines, since the case of a finite extension k'/k is somewhat simpler than the more general case treated in EGA IV₃, 11.9.10 b).

⁽²⁴⁾**N.D.E.** Specify this. . .

Suppose for simplicity U retrocompact in X (cf. EGA IV₃, 11.10.10 and 11.9.17 for the general case). The case of X locally noetherian is mentioned only for the record; it is contained in 4.3.

In the second case considered, one may evidently suppose S and X affine, so X and U are of finite presentation over $S = \operatorname{Spec}(A)$. The ring A is the inductive limit of its $\mathbf{Z}$ -subalgebras A_i of finite type. The “patented procedure” already used (cf. EGA IV₃, 8.8.2 and 8.10.5 (iii)) shows that there exists an i , an affine scheme X_i over $S_i = \operatorname{Spec}(A_i)$, and an open subset U_i in X_i , from which X, U are deduced by base change $S \rightarrow S_i$. Let E_i be the subset of S_i consisting of the $s \in S_i$ such that $(U_i)_s$ is schematically dense in $(X_i)_s$.

⁽²⁵⁾ By EGA IV₂, 5.9.9 and 5.10.2, E_i is the set of $s \in S_i$ such that $(U_i)_s$ contains the set $\operatorname{Ass} \mathcal{O}_{(X_i)_s}$ of points “associated” with the structure sheaf of $(X_i)_s$, and by EGA IV₃, 11.9.17.1 this condition is of constructible nature, i.e. E_i is a constructible subset of S_i .

⁽²⁵⁾ By 4.5, the inverse image of E_i by $S_j \rightarrow S_i$ (resp. by $S \rightarrow S_i$) is E_j (resp. the set E of $s \in S$ such that U_s is schematically dense in X_s). Moreover, by hypothesis $E = S$, which is also the inverse image by each $S \rightarrow S_i$ of S_i . By EGA IV₃, 8.3.11, this implies that there exists a $j \geq i$ such that $E_j = S_j$, i.e. such that for every $s \in S_j$, $(U_j)_s$ is schematically dense in $(X_j)_s$.

Then, by 4.4 applied to (S_j, X_j, U_j) and to the base change $S \rightarrow S_j$, it follows that U is schematically dense in X . One will note moreover that here one uses 4.4 only in the case of a family whose index set is finite (in fact reduced to one element), in which case the proof of 4.4 simplifies considerably, as the reader will verify for himself.

Theorem 4.7. *Let S be a prescheme, G a group of multiplicative type and of finite type over S . For every integer $n > 0$, let ${}_nG$ be the kernel of $n \cdot \operatorname{id}_G$. Then the family of subpreschemes $({}_nG)_{n>0}$ of G is schematically dense (def. 4.1).*

We distinguish two cases.

a) Case of S locally noetherian. Then by 4.3 one is reduced to the case where S is the spectrum of a field k . By 4.5.0, one may suppose k algebraically closed and G diagonalizable, i.e. of the form $D_k(M)$, where M is a commutative group of finite type. Then M is of the form $\Gamma \times \mathbf{Z}^r$, with Γ finite, so G is of the form $G_1 \times G_2$, with $G_1 = D(\Gamma)$ and $G_2 = \mathbf{G}_m^r$, and for n multiplicatively large (namely n a multiple of the order of Γ) one shall have ${}_nG = G_1 \times {}_nG_2$ since one shall have ${}_nG_1 = G_1$. Applying 4.3 again to the projection $G \rightarrow G_1$, one is reduced to the case of G_2 , i.e. to the case where $G = \mathbf{G}_m^r$. As G is then reduced, it amounts to the same thing to say that $({}_nG)_{n>0}$ is schematically dense in G or that the union of the ${}_nG$ is dense in G for the usual topology. Since ${}_nG = ({}_n\mathbf{G}_m)^r$, one is reduced to the case of $G = \mathbf{G}_m$, hence G irreducible of dimension 1. Then this follows from the fact that the union of the ${}_nG$ (equal to the set of roots of unity in k) is infinite.

b) General case. For every point s of S , there exists an open neighborhood U of s and a faithfully flat quasi-compact morphism $S' \rightarrow U$ such that $G' = G_{S'}$ is diagonalizable, i.e. of the form $D_{S'}(M)$. By 4.5.0 again, one may reduce to the case of G' , so one may suppose G diagonalizable, hence it comes from the absolute group $H = D_{\mathbf{Z}}(M)$ over $\operatorname{Spec}(\mathbf{Z})$ by base change $S \rightarrow \operatorname{Spec}(\mathbf{Z})$. By the proof of a), for every $s \in \operatorname{Spec}(\mathbf{Z})$ the family $({}_nH_s)_{n>0}$ is schematically dense in H ; it now suffices to apply 4.4.⁽²⁶⁾

Corollary 4.8. *a) Let $u, v : H \rightarrow G$ be two homomorphisms of S -group preschemes, with H of multiplicative type⁽²⁷⁾ and of finite type, and suppose that for every integer $n > 0$ the restrictions of u and v to ${}_nH$ are identical. Then $u = v$.*

⁽²⁵⁾N.D.E. The original has been expanded in what follows.

⁽²⁶⁾N.D.E. This result will be needed in X 4.3 for S not locally noetherian (namely $S = \operatorname{Spec}(\widehat{A} \otimes_A \widehat{A})$, where $\widehat{A}$ is the completion of the noetherian local ring A).

⁽²⁷⁾N.D.E. “Diagonalizable” has been replaced by “of multiplicative type”.

b) Let H_1, H_2 be two subgroups of multiplicative type and of finite type of a group prescheme G , and suppose that for every integer $n > 0$ one has ${}_nH_1 = {}_nH_2$; then $H_1 = H_2$.

The first assertion follows from the second by considering the subgroups H_1 and H_2 of $H \times G$, graphs of u and v . To prove b), let $H = H_1 \cap H_2$; it is a subgroup prescheme of H_i ($i = 1, 2$), and one must show that it is identical to H_i ; now the hypothesis means that it contains the ${}_nH_i$. One is therefore reduced to proving the

Corollary 4.9. *Let G be an S -group of multiplicative type and of finite type, and H a subgroup prescheme containing the ${}_nG$, $n > 0$. Then $H = G$.*

By 4.7 one is reduced to proving that the subprescheme H is closed, or again that it equals G set-theoretically. This reduces us to the case where S is the spectrum of a field, but then every subgroup prescheme of G is closed (VI_B 1.4.2), whence the conclusion.

Remark 4.10. Under the conditions of 4.7, let m be an integer > 0 with the following properties: for every $s \in S$, m is not a power of the characteristic of $\kappa(s)$, and if G_s is of type M , the prime divisors of the torsion of M divide m . (N.B. this second condition is always satisfied if G is a torus.) Then the proof given shows that in the statement of 4.7 and corollaries 4.8 and 4.9 one may restrict to considering the subgroups of the form $(m^r)G$, with $r > 0$.

Moreover, one sees immediately that when G is smooth over S , for every point $s \in S$ there exists an open neighborhood U of s and an integer $m > 0$ prime to all the residue characteristics of U , and satisfying the preceding conditions for G_U . (Take, for example, for m the order of the torsion of the type of G at s , cf. 1.4.1 a) and 2.1 e).) Under these conditions one finds $(m^r)G$ which are finite and étale over S , and whose family is schematically dense, provided one restricts over U . This remark allows one in certain cases (in particular those involving theorems 3.2 and 3.2 bis, cf. XI 6, but not those involving theorems 3.6 and 3.6 bis) to dispense with theorem 3.1, which involves Hochschild cohomology.

5. Central homomorphisms of groups of multiplicative type

Lemma 5.0. ⁽²⁸⁾ *Let $(A, \mathfrak{m})$ be a noetherian local ring, $S = \text{Spec}(A)$, and H a finite scheme over S , hence $H = \text{Spec}(B)$, where B is finite over A . Let K be a subscheme of H such that by reduction modulo $\mathfrak{m}^{n+1}$ one has $K_n = H_n$ for every n . Then $K = H$.*

Let s be the closed point of S . Let us first note that K is a closed subscheme of H . Indeed, it is a priori a closed subscheme of an open subset U of H . But K , and hence U , contains the fiber $K_s = H_s$; since the morphism $H \rightarrow S$ is finite, hence closed, it follows that the complement of U is empty, i.e. $U = H$. Thus K is defined by an ideal I of B . The hypothesis implies that I is contained in $\mathfrak{m}^n B$ for every n ; since B is a finite A -module, it is separated for the $\mathfrak{m}$ -adic topology, whence $I = 0$.

Theorem 5.1. *Let $u, v : H \rightarrow G$ be two homomorphisms of S -group preschemes, with H of multiplicative type and of finite type, and S locally noetherian or G of finite presentation over S . Let $s \in S$ be such that $u_s = v_s$, and suppose u_s central. Then there exists an open neighborhood U of s such that $u_U = v_U$.*

Let us distinguish the two cases:

a) S locally noetherian.⁽²⁹⁾ Let $K = \text{Ker}(u, v)$ be the inverse image of the diagonal of $G \times_S G$ by the morphism (u, v) ; it is a subgroup prescheme of H . We want to find U such that $K_U = H_U$. Note that, as S is locally noetherian and H of finite type over S , H is

⁽²⁸⁾**N.D.E.** This lemma has been made explicit; it is used several times in what follows (implicitly in the original).

⁽²⁹⁾**N.D.E.** The original has been expanded in what follows.

locally noetherian, so the immersion $K \hookrightarrow H$ is of finite type (cf. EGA I, 6.3.5). Thus K is of finite type over S , hence of finite presentation over S , since S is locally noetherian. Consequently, by EGA IV₃, 8.8.2.4, to show that there exists an open neighborhood U of s such that $K_U = H_U$, it suffices to show that $K_{S'} = H_{S'}$, where S' is the spectrum of $A = \mathcal{O}_{S,s}$. One may therefore suppose S local with closed point s . Replacing the noetherian local ring A by its completion $\hat{A}$ if necessary, which introduces a faithfully flat and quasi-compact base change $\hat{S} \rightarrow S$, one may even suppose⁽³⁰⁾ A complete if one wishes.

By 3.4 one has $u_n = v_n$ for every n , where, as usual, the subscript n indicates reduction modulo $\mathfrak{m}^{n+1}$ ($\mathfrak{m}$ being the maximal ideal of A). For every integer $m > 0$, denoting by ${}_m u, {}_m v$ the homomorphisms ${}_m H \rightarrow G$ induced by u, v , one therefore also has $({}_m u)_n = ({}_m v)_n$. This being true for every n , and ${}_m H$ being finite over S by 2.2, it follows that ${}_m u = {}_m v$ by 5.0. This being true for every m , one therefore has $u = v$ by 4.7.

b) G of finite presentation.⁽³¹⁾ Since H is also of finite presentation over S , by EGA IV₃, 8.8.2, we may again suppose S local with closed point s and prove that then $u = v$. If $f : S' \rightarrow S$ is a faithfully flat quasi-compact morphism, and if one denotes by f_H the morphism $H' \rightarrow H$ and by $u', v' : H' \rightarrow G'$ the morphisms deduced from u, v , then the equality $u' = v'$ implies $u' \circ f_H = v' \circ f_H$, whence $u = v$, since f_H is an epimorphism. Thus, making a faithfully flat and quasi-compact extension of the base if necessary, one may suppose in addition H diagonalizable, hence of the form $D_S(M)$, with M a commutative group of finite type. Introduce, as in the proof of 4.6, the increasing filtering family of $\mathbf{Z}$ -subalgebras of finite type A_i of $A = \mathcal{O}_{S,s}$, and $S_i = \text{Spec}(A_i)$.⁽³¹⁾ Note that $H = D_S(M)$ comes, for every i , from the diagonalizable group $H_i = D_{S_i}(M)$. Since G is of finite presentation over S , by EGA IV₃, 8.8.2 (see also VI_B, 10.2 and 10.3), there exists an index i , a group prescheme G_i of finite presentation over S_i , and morphisms of S_i -groups $u_i, v_i : H_i \rightarrow G_i$ from which u, v come by base change. Let s_i be the image of s in S_i , and let $\rho_i : H_i \times_{S_i} G_i \rightarrow G_i$ be the morphism of S_i -preschemes defined by $\rho_i(h, g) = u_i(h)gu_i(h)^{-1}$. Then, since u_s is central, $\rho_s = \rho_i \times_{\kappa(s_i)} \kappa(s)$ equals the second projection, and hence the same is true of ρ_i (since $\kappa(s_i) \rightarrow \kappa(s)$ is faithfully flat and quasi-compact), i.e. $(u_i)_{s_i}$ is central. Similarly, since $u_s = v_s$, one has $(u_i)_{s_i} = (v_i)_{s_i}$. One may then apply a) to the situation over S_i , whence immediately the announced conclusion.

Corollary 5.2. *Let $u : H \rightarrow G$ be a homomorphism of S -group preschemes, with H of multiplicative type and of finite type, and S locally noetherian or G of finite presentation over S . Let $s \in S$ and suppose that u_s is the unit homomorphism. Then there exists an open neighborhood U of s such that u_U is the unit homomorphism.*

Corollary 5.3. *Let u, H, G be as in 5.2, and suppose in addition G separated over S . Let U be the set of $s \in S$ such that u_s is the unit homomorphism. Then U is an open and closed subset of S , and $u_U : H_U \rightarrow G_U$ is the unit homomorphism.*

It remains only to prove that U is closed. Now let $K = \text{Ker}(u)$; since G is separated over S , K is a closed subscheme of H , and U is the set of $s \in S$ such that $K_s = H_s$. One then easily sees that U is closed, for example as an application of VIII 6.4 (H being essentially free over S by VIII 6.3), or by noting that one may suppose S reduced and U dense in S , hence schematically dense in S , which implies H_U schematically dense in H since H is flat over S ,⁽³²⁾ and as u and the unit homomorphism coincide on H_U , they coincide on H .

⁽³⁰⁾**N.D.E.** For $K = H$ if $K_{\hat{S}} = H_{\hat{S}}$, cf. SGA 1, VIII 5.4; but what follows does not use the hypothesis “ A complete”.

⁽³¹⁾**N.D.E.** The original has been expanded in what follows.

⁽³²⁾**N.D.E.** And of finite presentation over S , cf. EGA IV₃, 11.10.5 (ii) and 11.9.10 (ii).

Corollary 5.4. *Let S be a prescheme, H and G two S -groups, with H of multiplicative type and of finite type and G separated and of finite presentation over S , $\pi : S' \rightarrow S$ a⁽³³⁾ universal effective epimorphism (for example a faithfully flat and quasi-compact morphism, or a morphism admitting a section, cf. IV 1.12) with geometrically connected fibers.⁽³⁴⁾*

Let $u' : H' \rightarrow G'$ be a central homomorphism of the S' -groups deduced from H, G by the base change $S' \rightarrow S$. Then there exists a unique homomorphism $u : H \rightarrow G$ of S -groups such that $u \times_S S' = u'$. When S'/S admits a section g , u is the morphism deduced from u' by the base change $g : S \rightarrow S'$.

Since π is a universal effective epimorphism, the same is true of $H' \rightarrow H$, whence uniqueness of u , cf. the beginning of the proof of 5.1 b). If π admits a section g , then $u' = \pi^*(u)$ implies $u = g^*\pi^*(u) = g^*(u')$.

For the existence of u , one is reduced, by IV 2.3, to showing that the two homomorphisms $u'_1, u'_2 : H'' \rightarrow G''$ of S'' -groups deduced from u' by the two base changes $\text{pr}_1, \text{pr}_2 : S'' = S' \times_S S' \rightarrow S'$ are identical. Now they coincide on the diagonal of S'' ; more precisely, the inverse images of u'_1 and u'_2 by the diagonal morphism $S' \rightarrow S''$ are identical (being equal to u').⁽³⁵⁾ Since u'_1 and u'_2 are central, one may apply 5.3 to the morphism $u'_1(u'_2)^{-1}$. There therefore exists an open and closed subset U of S'' , containing the diagonal of S'' , such that u'_1 and u'_2 coincide over U .

Now the fibers of S'/S being geometrically connected, the same is true of those of S''/S , which are therefore a fortiori connected, whence it follows that U (containing the diagonals of the said fibers) contains the said fibers, and hence equals S'' , which completes the proof.

Corollary 5.5. *Let S be a prescheme, K an S -group prescheme locally of finite type⁽³⁶⁾ and with connected fibers, H an invariant subgroup of multiplicative type and of finite type of K . Then H is a central subgroup of K .*

⁽³⁷⁾ Let us first note that $\pi : K \rightarrow S$ has geometrically connected fibers, since for a group scheme locally of finite type over a field, connected implies geometrically connected (cf. VI_A 2.4). One may then apply 5.4 by putting $G = H$ and $S' = K$ to the homomorphism of K -groups $u' : H_K \rightarrow H_K$ defined set-theoretically by $(h, k) \mapsto (khk^{-1}, k)$, which is central since H is commutative. The inverse image of u' by the unit section $\varepsilon : S \rightarrow K$ is the identity homomorphism of K , so by 5.4 the same is true of $H_K \rightarrow H_K$, hence H is central in K . Let us state the variants of the preceding results for central subgroups of multiplicative type. One obtains, proceeding as for the preceding results (and using 3.2 bis):

Theorem 5.1 bis. *Let G be an S -group prescheme, H_1 and H_2 two subgroup preschemes of multiplicative type and of finite type, with $(H_1)_s$ central. Suppose S locally noetherian or G of finite presentation over S . Then for every $s \in S$ such that $(H_1)_s = (H_2)_s$ there exists an open neighborhood U of s such that $H_1|U = H_2|U$.*

Corollary 5.3 bis. *Under the preceding conditions, suppose in addition G separated over S . Let U be the set of $s \in S$ such that $(H_1)_s = (H_2)_s$. Then U is an open and closed subset of S , and $H_1|U = H_2|U$.*

⁽³³⁾**N.D.E.** “Covering morphism for the faithfully flat and quasi-compact topology” has been replaced by “universal effective epimorphism”, in order to be able to apply this to the morphism $K \rightarrow S$ of 5.5, and the beginning of the proof has been modified accordingly.

⁽³⁴⁾**N.D.E.** This is the case, for example, if $S = \text{Spec } k$ and $S' = \text{Spec } k'$, where k is a field and k' a radical extension of k , cf. X, Prop. 1.4.

⁽³⁵⁾**N.D.E.** The following sentence has been added.

⁽³⁶⁾**N.D.E.** The hypothesis “locally of finite type” has been added to agree with the reference VI_A, 2.4. In fact, one can show that over a field every connected group scheme is geometrically connected.

⁽³⁷⁾**N.D.E.** The original has been expanded in what follows.

Corollary 5.4 bis. *Let S be a prescheme, G a group prescheme separated and of finite presentation over S , $S' \rightarrow S$ a covering morphism for the faithfully flat quasi-compact topology, with geometrically connected fibers, H' a subgroup of multiplicative type and of finite type of $G' = G \times_S S'$.*

Then there exists a unique subgroup H of G such that $H' = H \times_S S'$, and H is of multiplicative type and of finite type over S , by 1.1 and 2.1 b). If $S' \rightarrow S$ admits a section g , H is the inverse image of the subgroup H' of G' by $g : S \rightarrow S'$.

Theorem 5.6. *Let S be a prescheme, $u : H \rightarrow G$ a homomorphism of S -group preschemes, with H of multiplicative type and of finite type, G of finite presentation over S .*

a) Suppose G to have connected fibers, and let $s \in S$ be such that $u_s : H_s \rightarrow G_s$ is a central homomorphism. Then there exists an open neighborhood U of s such that the homomorphism $H|_U \rightarrow G|_U$ induced by u is central.

b) Suppose that for every $s \in S$, $u_s : H_s \rightarrow G_s$ is central. Then u is central.

⁽³⁸⁾ Let us prove (a). Proceeding exactly as in the proof of 5.1 (b), one may suppose S local with closed point s , then $H = D_S(M)$ diagonalizable, then that there exists a $\mathbf{Z}$ -subalgebra of finite type A_i of $A = \mathcal{O}_{S,s}$ and a morphism $u_i : D_{S_i}(M) \rightarrow G_i$ such that u_{s_i} is central (where s_i is the image of s in S_i) and from which u comes by base change. This reduces us to the case where S is local noetherian with closed point s , and one must then prove that u is central.

Let K be the subprescheme $\text{Ker}(v, w)$, where $v, w : H \times_S G \rightrightarrows G$ are defined by

$$v(h, g) = g, \quad w(h, g) = \text{int}(u(h)) \cdot g = u(h)gu(h)^{-1}.$$

Then K is a sub- G -group of the G -group $H_G = H \times_S G$, and one wants to show that it equals H_G itself. By 4.9 one is reduced to proving that it contains the ${}_m(H_G) = ({}_m H) \times_S G$ for every integer $m > 0$, which reduces us to the case where $H = {}_m H$, hence H finite over S .

Let e be the unit element of the fiber G_s ; then $S' = \text{Spec}(\mathcal{O}_{G,e})$ is a noetherian local scheme (G being of finite presentation over noetherian S); put $S'_n = S' \times_S S_n$, where $S_n = \text{Spec}(A/\mathfrak{m}^{n+1})$. Then $K_{S'} = K \times_G S'$ is a subprescheme of $H_{S'} = H \times_S S'$, and by 3.9 one has $K_{S'_n} = H_{S'_n}$ for every n .⁽³⁹⁾ Since $H_{S'}$ is finite over S' , one concludes from 5.0 (applied to the noetherian local ring B) that $K_{S'} = H_{S'}$.

On the other hand, since $H \times_S G$ is noetherian (being of finite presentation over noetherian S), the immersion $K \hookrightarrow H \times_S G$ is of finite type (cf. EGA I, 6.3.5), so K is of finite type, hence of finite presentation over G . Then the equality $K_{S'} = H_G \times_G S'$ implies, by EGA IV₃, 8.8.2.4, that there exists an open neighborhood W of e in G such that $K \times_G W = H_G \times_G W = H \times_S W$. Thus K contains the open neighborhood $V = H \times_S W$ of the unit section of G_H over H .

For every $t \in H$, the fiber G_t (being an algebraic $\kappa(t)$ -group) is Cohen-Macaulay (VI_A, 1.1.1), and hence without embedded components; as it is in addition connected, hence irreducible (VI_A, 2.4), it has its generic point as its unique associated point. Thus by EGA IV₂, 3.1.8, the open subset V_t is schematically dense in G_t . By 4.3, V is therefore schematically dense in G_H . Moreover, since K is a sub- H -group of G_H , it induces on each fiber G_h a subgroup K_h , and as the latter contains an open neighborhood of the neutral element and G_h is connected, it follows that $K_h = G_h$, whence $K = G_H$ set-theoretically. Thus K is a closed subprescheme of G_H containing the schematically dense subprescheme V , hence $K = G_H$, which proves a).

Finally, b) is a direct consequence of 3.9, in view of the fact that to check that a subprescheme K of $P = G \times_S H$ is identical to P , it suffices to check that for every base change $S' \rightarrow S$, where S' is an artinian quotient of a local ring of S , one has $K' = P'$.

⁽³⁸⁾**N.D.E.** The original has been expanded in the proof of a).

⁽³⁹⁾**N.D.E.** The original has been expanded and corrected, replacing “unit section of $(G_H)_n$ over H_n ” by $H_{S'_n}$.

6. Monomorphisms of groups of multiplicative type, and canonical factorization of a homomorphism of such a group

Lemma 6.1. *Let S be a quasi-compact prescheme, G a commutative S -group prescheme, of finite presentation and quasi-finite over S . Then there exists an integer $n > 0$ such that $n \cdot \text{id}_G = 0$, i.e. $G = {}_n G$.*

If S is the spectrum of a field k , G is finite over k , and by VII_A 8.5 it suffices to take $n = \deg(G/k)$. Suppose now $S = \text{Spec}(A)$ local artinian; let k be the residue field of A , $G_0 = G \otimes_A k$, and $n_0 = \deg(G_0/k)$. Distinguish two cases.

a) k is of characteristic zero. Then G_0 is separable over k , hence G unramified over S . Then the unit section of G is an open immersion, hence ${}_n G = \text{Ker}(n \cdot \text{id}_G)$ is an open subscheme of G , so for it to equal G it suffices that it do so set-theoretically; one may therefore then take $n = n_0$.

b) k is of characteristic $p > 0$. Denote by $\mathfrak{m}$ the maximal ideal of A and put $S_r = \text{Spec}(A/\mathfrak{m}^{r+1})$ for every r . Let m be an integer such that $\mathfrak{m}^{m+1} = 0$; I say that one may take $n = p^m n_0$. Indeed, proceeding by induction on m and putting $n' = p^{m-1} n_0$, one may suppose that $n' \cdot \text{id}_G$ induces the zero endomorphism in $G_{m-1} = G \times_S S_{m-1}$.

⁽⁴⁰⁾ Denote by E the set of endomorphisms u of G having this property and by K the kernel of the morphism of group functors $G \rightarrow \prod_{S_{m-1}/S} G$ (cf. III 0.1.2). Then E is identified with $\text{Hom}_{S\text{-gr.}}(G, K)$; in particular the abelian group law on E is induced by that of K . Now by III 0.9, K is the S -group functor associating with every $g : T \rightarrow S$ the k -vector space

$$\text{Hom}_{\mathcal{O}_{T_0}}(g_0^*(\Omega_{G_0/S_0}^1), \mathfrak{m}^m \mathcal{O}_T);$$

one therefore has $pu = 0$ for every $u \in E$. Thus one has $pn' \cdot \text{id}_G = 0$, i.e. $p^m n_0 \cdot \text{id}_G = 0$.

Suppose now S noetherian (one reduces to this in 6.1 by the usual reduction to the noetherian case⁽⁴¹⁾). It then suffices to combine the preceding with the

Lemma 6.2. *Let X be a prescheme quasi-finite over noetherian S , and consider an increasing filtering family of subpreschemes $(X^i)_{i \in I}$ having the following property:*

for every $s \in S$ and $n \geq 0$, putting $S_{s,n} = \text{Spec}(\mathcal{O}_{S,s}/\mathfrak{m}^{n+1})$, there exists $i \in I$ such that $X^i \times_S S_{s,n} = X \times_S S_{s,n}$.

Under these conditions there exists an $i \in I$ such that $X^i = X$.

Since S is noetherian, there exists a maximal open subset U such that one has $X|U = X^i|U$ for large i ; one shall show that $U = S$. In other words, one shall show that if $U \neq S$, one can find a U' strictly larger than U , and an i such that $X|U' = X^i|U'$. Localizing at a maximal point s of $S - U$, one is reduced to the case where S is local with closed point s and $U = S - \{s\}$. (Indeed, if one denotes by $S' = \text{Spec}(\mathcal{O}_{S,s})$ and if there exists i such that $X \times_S S' = X^i \times_S S'$, then⁽⁴²⁾ there exists an open neighborhood V of s such that $X|V = X^i|V$; thus, taking i sufficiently large that $X^i|U = X|U$, one shall have $X|W = X^i|W$, where $W = U \cup V$.)

Then for large i , since $X|U = X^i|U$, one sees that X^i is a closed subprescheme of X defined by an ideal $\mathcal{J}^{(i)}$ with support contained in $X_s = X_0 = X \times_S S_0$ (where $S_n = \text{Spec}(A/\mathfrak{m}^{n+1})$, $S = \text{Spec}(A)$). As X_0 is quasi-finite over S_0 , X_0 is a finite closed subset of noetherian X ,

⁽⁴⁰⁾**N.D.E.** The original has been expanded in what follows, specifying the results of Exp. III that are used.

⁽⁴¹⁾**N.D.E.** Since S is quasi-compact, it is covered by finitely many affine open subsets, so one is reduced to the case where $S = \text{Spec}(A)$. Then, since G is of finite presentation over S , one may apply EGA IV₃, 8.8.2.

⁽⁴²⁾**N.D.E.** $X - \{s\}$ has been corrected above to $S - \{s\}$. On the other hand, recall that a quasi-finite morphism is supposed of finite type, cf. EGA II, 6.2.3 (and EGA III₂, Err_{III} 20 for the definition of “locally quasi-finite”). Thus here (S being noetherian), X is of finite presentation over S , each immersion $X^i \hookrightarrow X$ is of finite type (by EGA I, 6.3.5), so X^i is of finite presentation over S , and one may apply EGA IV₃, 8.8.2.

so $\mathcal{J}^{(i)}$ is a module of finite length. It follows that there exists an integer $n \geq 0$ such that $\mathcal{J}^{(i)} \cap \mathfrak{m}^{n+1}\mathcal{O}_X = 0$. On the other hand, by the hypothesis in 6.2 one may suppose (on increasing i) that the image of $\mathcal{J}^{(i)}$ in $\mathcal{O}_{X_n} = \mathcal{O}_X/\mathfrak{m}^{n+1}\mathcal{O}_X$ is zero. This implies $\mathcal{J}^{(i)} = 0$, hence $X^i = X$. Q.E.D.

Lemma 6.3. *Let S be a prescheme, K a group prescheme over S , locally of finite presentation, $s \in S$ such that K_s is quasi-finite (resp. unramified) over $\kappa(s)$ at the unit element. Then there exists an open neighborhood U of s such that $K|U$ is locally quasi-finite (resp. unramified) over U .⁽⁴³⁾*

Indeed, let V be the set of points x of K such that, denoting by t the image of x in S , the fiber K_t is quasi-finite (resp. unramified) over $\kappa(t)$ at x , i.e. such that x is isolated in K_t (resp. and its local ring in K_t a separable extension of $\kappa(t)$). One knows that V is open since K is locally of finite presentation over S ,⁽⁴⁴⁾ so if ε denotes the unit section of K , $\varepsilon^{-1}(V)$ is open. By hypothesis it contains s , so it is an open neighborhood U of s . The latter does the job; in other words $t \in U$ implies that K_t is locally quasi-finite (resp. unramified) over $\kappa(t)$: indeed, since K_t is a group locally of finite type over $\kappa(t)$, this follows from the fact that it is quasi-finite (resp. unramified) over $\kappa(t)$ at the point $\varepsilon(t)$, cf. VI_B 1.3.

Combining 6.1 and 6.3 one finds the

Theorem 6.4. *Let S be a prescheme, H an S -group of multiplicative type and of finite type, K a closed subgroup prescheme, of finite presentation over S . Let $s \in S$ be such that K_s is finite.*

Then there exists an open neighborhood U of s such that $K|U$ is contained in ${}_nH|U$ for some n , and a fortiori (${}_nH$ being finite over S) such that $K|U$ is finite over U .

Using Nakayama's lemma, one deduces:

Corollary 6.5. *With the preceding notation, suppose that K_s is the unit group. Then there exists an open neighborhood U of s such that $K|U$ is the unit group.*

Corollary 6.6. *Let $u : H \rightarrow G$ be a homomorphism of S -group preschemes, with H of multiplicative type and of finite type, and G separated over S . Suppose in addition S locally noetherian, or G locally of finite presentation⁽⁴⁵⁾ over S . Let $s \in S$ be such that $u_s : H_s \rightarrow G_s$ is a monomorphism; then there exists an open neighborhood U of s such that u induces a monomorphism $H|U \rightarrow G|U$.*

Indeed, let $K = \text{Ker}(u)$; the hypothesis on u_s means that K_s is the unit group, the conclusion that $K|U$ is the unit group for suitable U . Now G being separated over S , K is a closed subgroup of H , and in the case where one does not suppose S locally noetherian, but on the other hand G locally of finite presentation over S , K is locally of finite presentation over S ,⁽⁴⁵⁾ and hence of finite presentation over S since it is separated over S (H being so) and quasi-compact over S (being closed in H , which is quasi-compact over S). One may therefore apply 6.5, whence 6.6.

Remark 6.6.1.⁽⁴⁶⁾ Under the hypotheses of 6.6, one will note that when G is in addition affine over S (resp. of finite presentation over S), $H|U \rightarrow G|U$ is even a closed immersion, as pointed out in 2.5 (resp. 2.6).

⁽⁴³⁾Cf. VI_B 2.5 (ii) for a more general statement.

⁽⁴⁴⁾**N.D.E.** Cf. EGA IV₃, 13.1.4 and EGA IV₄, 17.4.1. Moreover, the image of x has been denoted by t (instead of s).

⁽⁴⁵⁾**N.D.E.** In fact it suffices to suppose G locally of finite type over S ; by EGA IV₄, 1.4.3 (v), this implies ($H \rightarrow S$ being of finite presentation) that $H \rightarrow G$ is locally of finite presentation, and hence also $K \rightarrow S$, which is deduced from it by base change.

⁽⁴⁶⁾**N.D.E.** The numbering 6.6.1 has been added, for later references.

Corollary 6.7. *Let $u : H \rightarrow G$ be a homomorphism of S -group preschemes, with H of multiplicative type and of finite type; suppose S locally noetherian or G locally of finite presentation over S .*

If all the homomorphisms induced on the fibers $u_s : H_s \rightarrow G_s$ are monomorphisms, then u is a monomorphism.

The reasoning is the same as in 6.6, the hypothesis that G is separated over S being unnecessary here to ensure that K is closed in H , since the hypothesis that the u_s are monomorphisms implies that K is set-theoretically reduced to the unit section of G .

Theorem 6.8. *Let $u : H \rightarrow G$ be a homomorphism of S -group preschemes, with H of multiplicative type and of finite type, and G separated over S . Suppose in addition S locally noetherian or G of finite presentation over S .*

Then $K = \text{Ker}(u)$ is a subgroup of multiplicative type and of finite type of H , and u factors as

$$H \xrightarrow{u'} H/K \xrightarrow{u''} G,$$

where H/K is of multiplicative type and of finite type, u' is the canonical homomorphism and is faithfully flat (and affine), and u'' is a monomorphism.

(N.B. As remarked in 6.6.1, u'' is a closed immersion if G is affine over S or of finite presentation over S .) It suffices to prove that K is of multiplicative type, the rest of the proposition then following from 2.7 and IV 5.2.6.

Suppose first G of finite presentation over S . This hypothesis being stable under base change, to prove that K is of multiplicative type one is reduced to the case where H is diagonalizable, i.e. $H = D_S(M)$, where M is a commutative group of finite type. Let $s \in S$; then K_s is a closed subgroup of $H_s = D_{\kappa(s)}(M)$, hence by 8.1⁽⁴⁷⁾ is of the form $D_{\kappa(s)}(N)$, where N is a quotient group of M . Put $K' = D_S(N)$; then K' is a subgroup of multiplicative type of H ; let $v' : K' \rightarrow G$ be induced by u .⁽⁴⁸⁾ Then v'_s is the unit homomorphism by construction, so by 5.2 there exists an open neighborhood U of s such that $v'|_U : K'|_U \rightarrow G|_U$ is the unit homomorphism. Thus, replacing S by U if necessary, one may suppose that v' is the unit homomorphism, and hence that u factors as

$$H \xrightarrow{u'} H/K' \xrightarrow{u''} G.$$

Now, since u''_s is deduced from u_s by factoring through $H_s \rightarrow H_s/\text{Ker}(u_s)$, u''_s is a monomorphism (IV 5.2.6), so by 6.6 there exists an open neighborhood U of s such that $u''|_U : (H/K')|_U \rightarrow G|_U$ is a monomorphism. Thus, restricting S if necessary, one sees that u'' is a monomorphism, hence $\text{Ker}(u) = \text{Ker}(u') = K'$, which proves that $\text{Ker } u$ is of multiplicative type.

The same proof is valid if, instead of supposing G of finite presentation over S , one supposes S locally noetherian, at least in the case where H is diagonalizable. In the case where one does not make this hypothesis on H , one must show that one can find a covering base change $S' \rightarrow S$, with S' locally noetherian, which trivializes H . This is indeed what one will see in the next exposé (X 4.6).

7. Algebraicity of formal homomorphisms into an affine group

Theorem 7.1. *Let A be a noetherian ring endowed with an ideal I such that A is separated and complete for the I -adic topology, $S = \text{Spec}(A)$, $S_n = \text{Spec}(A/I^{n+1})$, H, G S -group preschemes, with H of isotrivial multiplicative type and G affine.*

⁽⁴⁷⁾**N.D.E.** The erroneous reference to IV 4.7.5 has been corrected. Note that no. 8 of the present exposé is independent of nos. 2 to 7.

⁽⁴⁸⁾**N.D.E.** “Induced by G ” has been corrected to “induced by u ”.

Then the canonical map

$$(x) \quad \theta : \text{Hom}_{S\text{-gr.}}(H, G) \longrightarrow \varprojlim_n \text{Hom}_{S_n\text{-gr.}}(H_n, G_n)$$

is bijective (where H_n, G_n are the S_n -groups deduced from H, G by the base change $S_n \longrightarrow S$).

Suppose first H diagonalizable, hence of the form

$$H = \text{Spec } A(M),$$

where $B = A(M)$ is the group algebra of the commutative group M with coefficients in A . One also has $G = \text{Spec}(C)$, where C is an A -algebra endowed with a diagonal map (satisfying the known axioms). Then the homomorphisms of S -groups $H \longrightarrow G$ correspond bijectively to the homomorphisms of A -algebras $\varphi : C \longrightarrow B$ compatible with the diagonal maps, i.e. such that for every $f \in C$,

$$\Delta_H(\varphi(f)) = (\varphi \otimes \varphi)(\Delta_G(f))$$

where Δ_H and Δ_G are the diagonal maps. One has an analogous description for the homomorphisms of S_n -groups $H_n \longrightarrow G_n$, defined by certain homomorphisms of A_n -algebras $\varphi_n : C_n \longrightarrow B_n$ (where one puts $A_n = A/I^{n+1}$, $B_n = B \otimes_A A_n$, $C = C \otimes_A A_n$). Put

$$\widehat{B} = \varprojlim_n B_n \quad \text{and} \quad \widehat{C} = \varprojlim_n C_n;$$

then $\widehat{B} = \varprojlim_n A_n(M)$ is identified with a submodule of the product A^M (namely that consisting of the families $(a_m)_{m \in M}$ of elements of A tending to 0 (for the I -adic topology) along the filter of complements of finite subsets of M). This already implies that the canonical homomorphism $B \longrightarrow \widehat{B}$ is injective,⁽⁴⁹⁾ since the projective system $\theta(\varphi) = (\varphi_n)_{n \geq 0}$ defines a homomorphism $\widehat{\varphi} = \widehat{C} \longrightarrow \widehat{B}$ such that the diagram below is commutative:

$$\begin{array}{ccc} C & \xrightarrow{\varphi} & B \\ \downarrow & & \downarrow \\ \widehat{C} & \xrightarrow{\widehat{\varphi}} & \widehat{B}. \end{array}$$

Let us prove that θ is surjective: take a projective system $(\varphi_n)_{n \geq 0}$ and prove that it comes by reduction from a φ in the left-hand side. A priori, (φ_n) defines a homomorphism on the completed algebras

$$\widehat{\varphi} : \widehat{C} \longrightarrow \widehat{B},$$

and everything amounts to seeing that its composite $\Phi : C \longrightarrow \widehat{C} \xrightarrow{\widehat{\varphi}} \widehat{B}$ with the canonical homomorphism $C \longrightarrow \widehat{C}$ maps C into B . Indeed, if this is so, one finds a homomorphism of A -algebras $\varphi : C \longrightarrow B$ reducing to the φ_n , whence one concludes immediately that it is compatible with the diagonal maps (since the φ_n are so and $B \otimes_A B \longrightarrow \widehat{B \otimes_A B}$ is injective, as one sees as above on replacing M by $M \times M$).

Let us note that the diagonal homomorphism Δ_H of H defines, on passing to completions, a homomorphism

$$\widehat{\Delta}_H : \widehat{B} \longrightarrow \widehat{B \otimes_A B},$$

and one has a commutative diagram (deduced by passing to the projective limit from the analogous diagrams defined by the φ_n):

$$\begin{array}{ccc} C & \xrightarrow{\Phi} & \widehat{B} \\ \Delta_G \downarrow & & \downarrow \widehat{\Delta}_H \\ C \otimes_A C & \xrightarrow{\Psi} & \widehat{B \otimes_A B}, \end{array}$$

⁽⁴⁹⁾**N.D.E.** What follows has been expanded.

where Ψ is the composite

$$C \otimes_A C \xrightarrow{\Phi \otimes \Phi} \widehat{B} \otimes_A \widehat{B} \longrightarrow \widehat{B \otimes_A B}$$

(the last arrow is the evident canonical homomorphism). It follows that for every $f \in C$, $\Phi(f)$ is an element of $\widehat{B}$ whose image under $\widehat{\Delta}_H$ is a “decomposable” element of $\widehat{B \otimes_A B}$, i.e. is in the image of $\widehat{B} \otimes_A \widehat{B}$.

Denote by $(e_m)_{m \in M}$ the canonical basis of $A(M)$ and by $(e_{m,m'})$ that of $A(M \times M) = A(M) \otimes_A A(M)$. Since $\Delta_H(e_m) = e_m \otimes e_m = e_{m,m}$ for every m , it now suffices to apply the

Lemma 7.2. *Let A be a noetherian ring, M a set, $(a_{m,m'})$ a family of elements of A indexed by $M \times M$, such that*

(i) $a_{m,m'} = 0$ if $m \neq m'$ (i.e. the support of the family is contained in the diagonal of $M \times M$),

(ii) One has

$$a_{m,m'} = \sum_i b_m^i c_{m'}^i$$

where the b^i, c^i are finitely many elements of A^M (i.e. $(a_{m,m'})$ belongs to the image of the canonical homomorphism $A^M \otimes_A A^M \longrightarrow A^{M \times M}$).

Under these conditions, the support of the family $(a_{m,m'})$ is finite.

By (i), the family $(a_{m,m'})$ is determined by knowing the $a_m = a_{m,m}$. Put, for every $x = (x_n)_{n \in M} \in A^{(M)}$:

$$(u \cdot x)_m = \sum_{m'} a_{m,m'} x_{m'},$$

i.e. interpret $(a_{m,m'})$ as the matrix of a homomorphism $u : A^{(M)} \longrightarrow A^M$. Then by (i) one simply has

$$(u \cdot x)_m = a_m x_m.$$

On the other hand, by (ii) one has

$$u \cdot x \in \sum_i A \cdot b^i,$$

so $u(A^{(M)})$ stays in an A -module of finite type. Consequently, denoting by $(e^m)_{m \in M}$ the canonical basis of $A^{(M)} \subseteq A^M$, the $a_m e^m$ stay in an A -module of finite type. As A is noetherian, the module they generate is itself of finite type, which implies (since the e^m are linearly independent) that all but finitely many of the a_m are zero. This proves 7.2 and hence 7.1 in the case where H is diagonalizable.

Let us now prove the general case of 7.1, where one merely supposes H isotrivial, i.e. that there exists a finite surjective étale morphism $S' \longrightarrow S$ such that $H' = H \times_S S'$ is diagonalizable. We shall use only the fact that $S' \longrightarrow S$ is finite and covering (for the faithfully flat and quasi-compact topology, or simply for the canonical topology of (Sch)) — thus the hypothesis “étale” could be replaced by “flat”.

Let $S'' = S' \times_S S'$; introduce similarly S'_n and $S''_n = S'' \times_S S_n = S'_n \times_{S_n} S'_n$, and $H', G', H'', G'', H'_n, G'_n, H''_n, G''_n$ deduced from H and G by the base changes which one can guess. Note that H' and H'' are now diagonalizable. Note also that S' , and hence S'' , is affine, and that if $S' = \text{Spec}(A')$, $S'' = \text{Spec}(A'')$, then A' and A'' are separated and complete for the topology defined by IA' resp. by IA'' (since A' and A'' are finite over A). Since $S' \longrightarrow S$ and $S'_n \longrightarrow S_n$ are covering, one obtains a commutative diagram of maps of sets whose two rows are exact:

$$\begin{array}{ccccc} \text{Hom}_{S\text{-gr.}}(H, G) & \longrightarrow & \text{Hom}_{S'\text{-gr.}}(H', G') & \rightrightarrows & \text{Hom}_{S''\text{-gr.}}(H'', G'') \\ \downarrow u & & \downarrow u' & & \downarrow u'' \\ \varprojlim_n \text{Hom}_{S_n\text{-gr.}}(H_n, G_n) & \longrightarrow & \varprojlim_n \text{Hom}_{S'_n\text{-gr.}}(H'_n, G'_n) & \rightrightarrows & \varprojlim_n \text{Hom}_{S''_n\text{-gr.}}(H''_n, G''_n). \end{array}$$

(N.B. The second row is exact as a projective limit of exact diagrams relative to the various $S'_n \rightarrow S_n$.) By what has already been proved (diagonalizable case), u' and u'' are bijective. The same is therefore true of u , which completes the proof of 7.1.

Corollary 7.3. *Under the conditions of 7.1, suppose in addition G smooth over S , and let $u_0 : H_0 \rightarrow G_0$ be a homomorphism of S_0 -groups. Then there exists a homomorphism of S -groups $u : H \rightarrow G$ lifting u_0 . Two such liftings u, u' are conjugate by an element g of $G(S)$ reducing to the unit element of $G_0(S_0)$.*

Follows from the conjunction of 3.6 and 7.1. To construct a u , one constructs the u_n successively, which is possible by 3.6, and by 7.1 the system (u_n) comes from a u . Given two liftings u and u' , to construct g such that $u' = \text{int}(g)u$, $g_0 = 1$, one constructs successively g_n such that $g_0 = 1$, g_n is deduced from g_{n+1} by reduction, $u'_n = \text{int}(g_n)u_n$; this is possible thanks to 3.6. Since A is separated and complete, the g_n come from a $g \in G(S)$, and to prove that $u' = \text{int}(g)u$ it suffices to use injectivity in assertion 7.1.

Remark 7.4. One will compare 7.1 with EGA III 5.4.1, which implies that statement 7.1 is valid if, instead of supposing H of multiplicative type and G affine, one supposes H proper over S , and G separated and locally of finite type over S . Having a statement like 7.1 without a properness hypothesis is rather exceptional, and must here be interpreted as one of the aspects of the great “rigidity” of the structure of a group of multiplicative type. The analogous statement with $G = H = \mathbf{G}_a$ (additive group) is false in general, as one sees by taking A of characteristic $p > 0$ and defining the u_n by reduction modulo I^{n+1} from an additive formal series

$$u(T) = \sum_{n \geq 0} a_n T^{p^n},$$

where the a_n are elements of A tending to 0 for the I -adic topology, but not for the discrete topology. Statement 7.1 also becomes false if one suppresses the hypothesis G affine, even for $H = \mathbf{G}_m$ (multiplicative group); one sees an example (with A a complete discrete valuation ring) by starting with an elliptic curve over the field of fractions K of A which reduces (in the Néron-Kodaira reduction theory, say) to the group $\mathbf{G}_m$ over the residue field k .⁽⁵⁰⁾ one shall therefore have a smooth commutative group scheme G over S whose special fiber is $\mathbf{G}_{m,k}$ (which allows one, thanks to 3.6, to define a projective system of isomorphisms $u_n : H_n \xrightarrow{\sim} G_n$, where $H = \mathbf{G}_{m,A}$), but whose generic fiber is an abelian variety, so that there is no homomorphism of S -groups $H \rightarrow G$ other than 0.

8. Subgroups, quotient groups and extensions of groups of multiplicative type over a field

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Scholium 8.0. ⁽⁵²⁾ *Let k be a field, H an affine k -group scheme. Denote by $k[H]$ its affine algebra and by $X(H) = \text{Hom}_{k\text{-gr.}}(H, \mathbf{G}_{m,k})$ the group of characters of H . By the independence lemma for characters, $X(H)$ is a linearly independent subset of $k[H]$. It follows that H is diagonalizable if and only if $X(H)$ generates $k[H]$ as a k -vector space.*

Proposition 8.1. *Let G be a diagonalizable (resp. multiplicative type) group scheme over a field k .⁽⁵³⁾ Then for every subgroup scheme H of G , H and G/H are diagonalizable (resp. of multiplicative type).*

⁽⁵⁰⁾**N.D.E.** One could make such an example explicit. . .

⁽⁵¹⁾Added in July 1969. This afterthought no. is independent of nos. 3 to 7.

⁽⁵²⁾**N.D.E.** This scholium has been added.

⁽⁵³⁾**N.D.E.** “Over a field k ”, implicit in the original, has been added; on the other hand, “algebraic group” has been replaced by “group scheme”, since G is not supposed of finite type.

Definition 1.1 reduces us immediately to the case without “resp.” An easy passage to the limit, using VI_B 11.13 and VIII 3.1, reduces us to the case where G is of finite type over k .⁽⁵⁴⁾ By VI_B 11.16,⁽⁵⁵⁾ one can find a finite family of nonzero elements

$$(8.1.1) \quad f_i = \sum a_{im} m, \quad a_{im} \in k$$

of the affine ring $k(M)$ of $G = D_k(M)$ such that the points of H (with values in an arbitrary k -algebra k') are the points $g \in G(k')$ such that one has

$$(8.1.2) \quad \tau_g f_i = \lambda_i(g) f_i, \quad \text{with } \lambda_i(g) \in k',$$

where τ_g denotes translation by g . Now one has

$$(8.1.3) \quad \tau_g f_i = \sum a_{im} \chi_m(g) m,$$

$\chi_m : G \rightarrow \mathbf{G}_{m,k}$ being the character associated with m , so that (8.1.2) is equivalent to the relation

$$(8.1.4) \quad \chi_m(g) = \chi_{m'}(g) \quad \text{if } m, m' \in Z_i,$$

Z_i denoting the set of $m \in M$ such that $a_{im} \neq 0$. This relation may also be written

$$\chi_{m'-m}(g) = 1 \quad \text{if } m, m' \in Z_i.$$

Denoting by N the subgroup of M generated by all the $m' - m$ (i variable, and $m, m' \in Z_i$), one deduces from the definition of $D_k(M)$ ($\simeq G$) that H is identified with $D_k(M/N)$. It then follows from VIII 3.1 that G/H is identified with $D_k(N)$. Q.E.D.

Proposition 8.2. ⁽⁵⁶⁾ *Let k be a field, H, K k -group schemes of multiplicative type and of finite type, and G a k -group scheme such that one has an exact sequence*

$$1 \rightarrow H \rightarrow G \rightarrow K \rightarrow 1$$

(which implies that G is of finite type over k).

- (i) *If one supposes G commutative or K connected, then G is of multiplicative type.*
- (ii) *If K and H are diagonalizable, K being a torus, then G is diagonalizable.*

For the proof of (i), the reader will refer to XVII 7.1.1, of which 8.2 (i) is a particular case; the case of a field is treated in part (i) of the proof of XVII 7.1.1 (not using the results of the following exposés).

(ii) Suppose now K and H diagonalizable:

$$K \simeq D_k(M) \quad \text{and} \quad H \simeq D_k(N),$$

⁽⁵⁶⁾ and suppose G of multiplicative type (which is the case by (i) if K is a torus). Then, by X 1.4, G is isotrivial over k , i.e. there exists a finite separable extension k'/k such that $G' = G \times_k k'$ is diagonalizable, hence $G' = D_{k'}(E)$ for a commutative group E , and one has an exact sequence

$$(1) \quad 0 \rightarrow D_{k'}(N) \rightarrow D_{k'}(E) \rightarrow D_{k'}(M) \rightarrow 0.$$

Thus, by VIII, 3.1 and 3.2, M is a subgroup of E , and one has an exact sequence

$$(2) \quad 0 \rightarrow M \rightarrow E \rightarrow N \rightarrow 0.$$

For a given extension E_0 of N by M , consider the diagonalizable k -group $G_0 = D_k(E_0)$; then the k -group functor A of automorphisms of the extension

$$(3) \quad 1 \rightarrow H \rightarrow G_0 \rightarrow K \rightarrow 1$$

⁽⁵⁴⁾**N.D.E.** Expand this “passage to the limit”: this uses 8.0 and also the equality $G/H = \varprojlim_i G_i/H_i$, cf. VI_B, proof of 11.17. To see that $H = \varprojlim_i (H \cap G_i)$, does one use the fact that H is closed in G ? Using this, one can give a direct proof. . .

⁽⁵⁵⁾**N.D.E.** VI_B 11.16 ought doubtless to be rewritten in the usual, more pleasant form. In particular, a single f_i suffices below. . .

⁽⁵⁶⁾**N.D.E.** The statement, as well as its proof, has been expanded.

i.e. the group subfunctor of $\underline{\mathrm{Aut}}_{k\text{-gr.}}(G_0)$ whose points over a k -prescheme T are the $\phi \in \mathrm{Aut}_{T\text{-gr.}}(G_T)$ inducing the identity on H_T and on K_T , is identified with $\underline{\mathrm{Hom}}_{k\text{-gr.}}(K, H)$, which by VIII 1.5 is the constant k -group with value $L = \mathrm{Hom}_{\mathrm{gr.}}(N, M)$.

One therefore sees that the classification of extensions G of K by H which over a separable closure k_s of k become isomorphic to extension (3) is the same as that of k -torsors for the étale topology under the constant group L_k . Denoting by $\Gamma = \mathrm{Gal}(k_s/k)$, these are classified by the Galois cohomology group (L being a trivial Γ -module):

$$H_{\mathrm{\acute{e}t}}^1(k, L_k) = H^1(\Gamma, L) = \mathrm{Hom}_{\mathrm{gr. top.}}(\Gamma, L).$$

If, in addition, K is a torus, M is torsion-free, and hence so is L , whence $\mathrm{Hom}_{\mathrm{gr. top.}}(\Gamma, L) = 0$; it follows that every extension of K by H is already diagonalizable.

Remark 8.3. As already pointed out in the proof of 8.2, the first assertion is stated and proved over an arbitrary base in XVII 7.1.1. On the other hand, the second assertion generalizes, with essentially the same proof, to the case of a normal integral base scheme (or more generally a geometrically unibranch one), using X 5.13 below.

Finally, one may also consider 8.1 as a corollary of the (considerably less trivial) result 6.8, also valid over an arbitrary base scheme.⁽⁵⁷⁾

⁽⁵⁷⁾**N.D.E.** Note, however, that the proof of 6.8 uses 8.1!

Bibliography

(58)

[BE_{ns}] N. Bourbaki, *Théorie des ensembles*, Chap. I–IV, Hermann, 1970.

⁽⁵⁸⁾**N.D.E.** Additional references cited in this Exposé.